

Honeywell

TotalPlant Solution (TPS) System

Remote GUS User's Guide

GU2C-400

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03/2007

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About This Document

This document describes how to install and configure Remote GUS and Remote GUS security and how to setup and configure Remote IKB Client.

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References

The following list identifies all documents that may be sources of reference for material discussed in this publication.

Document Title
TPS System Implementation Guide, TP08X
Carbon Copy Access Edition Version 5.5 User's Guide

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World Wide Web

The following lists Honeywell's World Wide Web sites that will be of interest to our industrial automation and control customers.

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Contents

1. Introduction

1.1 Remote GUS (Carbon Copy AE) Overview

General Description

Carbon Copy® Access Edition (AE) is a stand-alone, remote control software package developed by Compaq® Computer Corporation and currently owned by Altiris®, Inc.

Carbon Copy AE allows a user on a local computer to connect to a remote computer. The computer initiating the connection and launching the Carbon Copy AE Remote Control utility is the Guest or Client (or local computer) and the computer accepting the connection is the Host (or remote computer). The Carbon Copy AE Remote Control utility allows the Client user to operate the Host computer as if sitting in front of it.

Honeywell provides Carbon Copy Access Edition (AE) to enable a user to access a TPS node, typically a GUS node, from a remote location. The implementation of Carbon Copy AE for GUS nodes is called Remote GUS.



REFERENCE - EXTERNAL

The Carbon Copy Access Edition Version 5.5 User's Guide is available as an installable component from the Honeywell Carbon Copy AE CD-ROM.

Host-Only and Full Versions

Honeywell provides two versions of Carbon Copy AE:

The **Host-only version** is a limited version that can function as a host, but not as a client. This means that Carbon Copy AE calls/connections can be received, but not initiated.

The **Full version** of Carbon Copy AE allows a PC to connect to a host, and allows the PC to be a host. This means that Carbon Copy AE calls/connections can be received as well as initiated

1 Introduction

1.1 Remote GUS (Carbon Copy AE) Overview



ATTENTION

- In this document, the Carbon Copy AE host is referred to as the **GUS Server (Host)** and the Carbon Copy AE client is referred to as the **GUS Client**.
 - The term "GUS Server" does not mean that the GUS node requires the Windows Server operating system.
 - Honeywell envisions that a TPS node, typically, will be configured only as a Host, regardless of whether the Full or Host-Only version is installed.
-

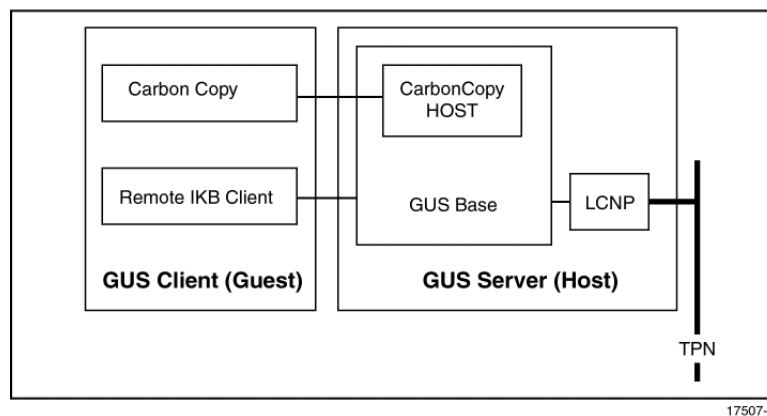
1.2 Remote GUS operation

Remote GUS block diagram

When using Carbon Copy AE, a remote user shares the resources of the TPS node with the local user. Because both a remote user and a local user can operate the mouse and the keyboard on a GUS, control is determined (by default) on a “first come, first served” basis.

The *guest components* (present on the GUS Client) define and initiate connections over a LAN or a phone line. The *host components* (present on the GUS Server) accept and service the connection after applying the necessary security checks.

Remote GUS Operation



Remote GUS connection scenario

After a connection is made, Carbon Copy AE displays the Gus Server (Host) screen on the GUS Client screen (including the Native Window).

Carbon Copy AE also forwards GUS Client mouse and keyboard events to the GUS Server (Host), and propagates Integrated Keyboard (IKB) and Operator Entry Panel (OEP) functional keys to the GUS Server (Host).

If installed, the **Remote IKB Client** software package provides current LED status and horn annunciation at the GUS Client. It also sends keylock position changes from the GUS Client to the GUS Server (Host).

1 Introduction

1.3 Remote GUS Hardware and Software

1.3 Remote GUS Hardware and Software

Remote GUS Hardware Requirements

Platform	CPU	RAM	Dialup or LAN	Transport	Video
GUS Server (See Note 1)	>= 300MHz	>= 64MB	10/100 Base-T LAN	TCP/IP	Single screen 1024 by 768
GUS Client	>= 300MHz	>= 64MB	56K modems are supported	TCP/IP	Single/dual screen >= 1024 by 768

⇒ **Note 1:** Installed software packages on the GUS Server (Host) may impose additional requirements, especially in the case of installed RAM.

Remote GUS Hardware Options

Platform	Optional
GUS Client	If GUS Client is a GUS node, these hardware options are supported: • Carroll Touch Screen at the GUS Client • Operator Entry Panel (OEP) at the GUS Client • Integrated Keyboard (IKB) at the GUS Client.

Remote GUS Software Requirements and Options

Platform	Mandatory	Optional
GUS Server (See Note 2)	Networked GUS Node Carbon Copy AE	SafeView GUS Display Server GUS Multiple Displays (See Note 3)
GUS Client	Carbon Copy AE Windows 2000 or Windows 2003/XP	If GUS Client is a GUS node, these software options are supported: • SafeView • Remote IKB Client • Carroll Touch Screen

⇒ **Note 2:** On GUS releases prior to R331, GUS Base software included Carbon Copy host computer components, which allowed host functionality to be added to those GUS nodes.

⇒ **Note 3:** Any GUS software packages supported in the TPN-connected GUS Node are optional. Refer to TPS System Software Package Installation in the *TPS System Implementation Guide*.

2. Instructions for Initial Remote GUS Setup

2.1 Introduction

The purpose of this section is to establish a simple connection to Carbon Copy AE using no security. If greater security is desired, then that can be established at a later time. The goal of the procedures in this section are to establish a simple connection that is stable and reliable.

2.2 Modem Installation (Optional)

Modem Type

Honeywell does not specify the modem for remote connections, but Honeywell Phoenix TAC recommends the US Robotics 56.6 external modem. Internal modems can be used, but they may be more difficult to establish, configure, and setup. They also may run a higher incompatibility potential with the Microsoft operating system than external modems.

One point to consider when purchasing the modem is that most modems do not come with the serial cable required to connect the modem to the PC. Unless specifically stated on the box containing the modem that there is a cable included, a cable will need to be purchased along with the modem.

Modem installation guidelines

When adding the modem to the PC, configure it as a standard modem.

Do not use any of the drivers that may be associated with the modem of choice; including drivers that came with the modem on the CD-ROM (if a CD-ROM accompanied the modem), or drivers available from the vendor's web site. Certain upgraded capabilities of the modem may confuse the connection and impact the success of the connection attempt.



ATTENTION

Perform the following procedure only if a modem will be used to service Carbon Copy AE connections.

2 Instructions for Initial Remote GUS Setup

2.2 Modem Installation (Optional)

Modem installation procedure - Using Windows 2000

Step	Action
1	Install the modem to either COM1 or COM2, power it on, and then reboot the node.
2	Windows 2000 has the ability to automatically detect and install new hardware. It should just install the modem, or it may inform you it has detected new hardware, do you want to install it? Click Yes .
3	Go to Start>Settings>Control Panel .
4	Select and double-click Phone and Modems Options .
5	Select the Modems tab.
6	Click the Properties button
7	Use the down arrow in the Maximum Port Speed select 115200 and click OK .
8	Select OK to exit
9	Continue to section 2.3 GUS Server (Host) Windows Group and User Configuration.

Modem installation procedure - Using Windows 2003/XP

Step	Action
1	Install the modem to either COM1 or COM2, power it on, and then reboot the node.
2	Windows 2003/XP has the ability to automatically detect and install new hardware. It should just install the modem, or it may inform you it has detected new hardware, do you want to install it? Click Yes .
3	Go to Start>Control Panel .
4	Select Printers and Other Hardware .
5	Select Phone and Modems Options .

2 Instructions for Initial Remote GUS Setup

2.2 Modem Installation (Optional)

Step	Action
6	Select the Modems tab.
7	Click the Properties button.
8	Select the Modems tab.
8	Use the down arrow in the Maximum Port Speed select 115200 and click OK .
9	Select OK to exit
10	Continue to section 2.3 GUS Server (Host) Windows Group and User Configuration.

2.3 GUS Server (Host) Windows Group and User Configuration for TAC Remote Diagnostics

Purpose

The following procedure describes how to create the Remote Diagnostics group for the TAC user account, how to create the TAC user account, and how to configure its group membership (remove it from the Users group and add it to the Administrators group and Remote Diagnostics group).

Procedure

Step	Action
Create a Remote Diagnostics Group	
1	At the GUS where Carbon Copy AE will be installed, logon as Administrator
2	Open Computer Management . (Right-click on the My Computer desktop icon and select Manage.) Expand Local Users and Groups .
3	Right-click on the Groups folder and select New Group . Create the following local group: Remote Diagnostics
4	Enter the following description: For access by the Carbon Copy Guest. Click the Create button.
5	Click Close to close the 'New Group' dialog box.
Create a TAC User Account	
6	Right-click on the Users folder and select New User . Create a following local user: TAC
7	Enter the following Full Name : Technical Assistance Center
8	Enter the following description: For Carbon Copy access by the remote expert.

2 Instructions for Initial Remote GUS Setup

2.3 GUS Server (Host) Windows Group and User Configuration for TAC Remote Diagnostics

Step	Action
9	Enter a password and confirmation. For example: TAC . Write down the password. _____
10	Uncheck the User Must Change Password at next Logon box .
11	Check the User Cannot Change Password box .
12	Check the Password Never Expires box .
13	Uncheck the Account Disabled box .
14	Click the Create button to create the new user.
15	Click the Close button to close the 'New User dialog box.
Configure Group Membership for TAC User Account	
16	Under Local Users and Groups , click the Users folder.
17	Double-click the new username (TAC).
18	Select the Member of tab and click the Add button. NOTE: Be sure this machine is indicated in the Look in port at the top of the Select Groups screen.
19	Select Administrators , click the Add button, and then click the OK button.
20	Click the Add button, select the Remote Diagnostics group, click the Add button, and then click OK .
21	Select Users from the Member of list box, click the Remove button, and then click the OK button.
22	Continue to the section 2.4 Carbon Copy AE Software Installation.

2.4 Carbon Copy AE Software Installation

Installation media

The **Host-Only version** of Carbon Copy AE is installed from the Honeywell Carbon Copy AE Host Only Version CD-ROM.

The **Full version** of Carbon Copy AE is installed from the Honeywell Carbon Copy AE Full Version CD-ROM

Software installation considerations



ATTENTION

Although the Full version of Carbon Copy can be loaded to any node, Honeywell recommends that it be installed only to non-TPS nodes.



ATTENTION

The following considerations apply to GUS R330 and APP R211 (TPS R130.1) and earlier:

- The Host-Only version of Carbon Copy is automatically installed from Honeywell TPS System Software CD-ROM with the following Honeywell TPS software packages:
 - APP Base Software
 - GUS Base Software
 - TPS Open Layer Software
 - If licensed, the Full version of Carbon Copy appears as a separate installable TPS package named "Carbon Copy."
-

Carbon Copy AE Installation Procedure

Step	Action
1	Insert the appropriate Carbon Copy AE CD-ROM . RESULT: A dialog box displays the installation directory.
2	Select Carbon Copy AE . RESULT: A dialog box displays the installation options. The default options are to install Carbon Copy and not the user's guide or the Adobe Acrobat Reader.
3	Click the OK button to proceed. RESULT: A dialog box displays the default install directory location.
4	Click the OK button to accept the default or type a new location. RESULT: Files are copied to the system and then a dialog box asks if the readme.txt file should be displayed.
5	Click No to skip the readme.txt Or Click Yes to launch Notepad to read the file. When you are done, close Notepad to finish the Carbon Copy install.
6	A dialog box indicates that the system should be restarted to complete the install. Click the YES button to restart the node.
7	For a GUS Server (Host), continue to section 2.5 Verify Startup of Carbon Copy Service on GUS Server. For a GUS Client, continue to section 2.6 Configure Carbon Copy AE.

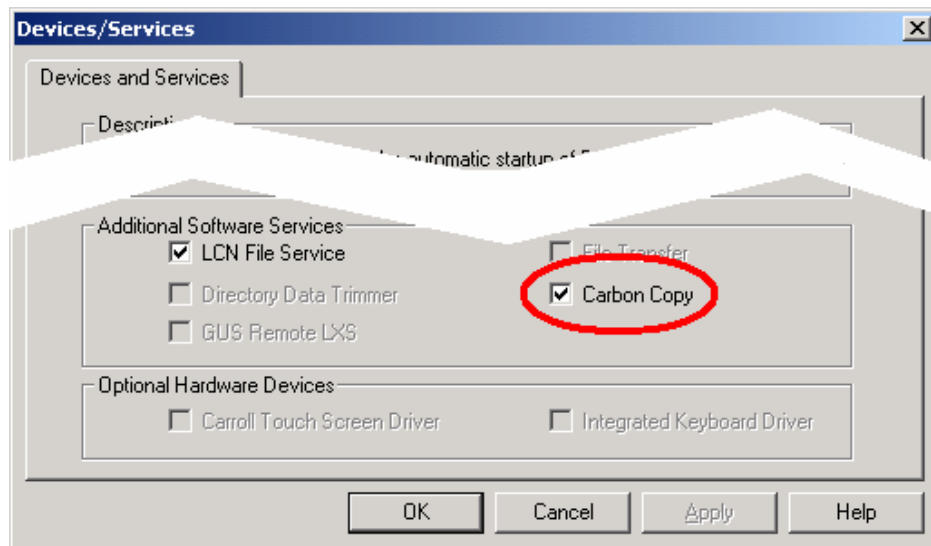
2.5 Verify Startup of Carbon Copy Service on GUS Server

Purpose

Perform the following procedure to verify that the Carbon Copy AE software has been set to startup as a service on the GUS Server node.

Procedure

Step	Action
1	Open the Configuration Utility by selecting: Start>Programs>Honeywell TPS>Configuration Utility
2	Select Configure>Devices/Services . RESULT: The Devices/Services dialog box opens.
3	See if Carbon Copy is checked in the Additional Software Services area.



- | | |
|---|---|
| 4 | If not checked, continue to step 5.

If checked, click Cancel, exit the Configuration Utility and go to step 8. |
| 5 | Check the box as shown in the illustration above, and then click OK and exit the Configuration Utility. |
| 6 | Shutdown and Restart the computer. |

2 Instructions for Initial Remote GUS Setup
2.5 Verify Startup of Carbon Copy Service on GUS Server

Step	Action
7	Logon as Administrator.
8	If not already loaded, load the Native Window with the GUS personality.
9	Continue to section 2.6 Configure Carbon Copy AE.

2.6 Configure Carbon Copy AE

Purpose

Carbon Copy AE has several features that can be set or restricted, depending upon the way it will be used. The settings in the following procedure are suggested initial entries and may be different for a particular site.

Procedure

Step	Action
1	Call up the Carbon Copy AE user interface (launch the program) by selecting: Start>Programs>Carbon Copy Access Edition>Carbon Copy
2	Select Options>Security . Select the Logins tab.
3	Select Require Logins . Select Use Native NT/2000 Security . Click OK .

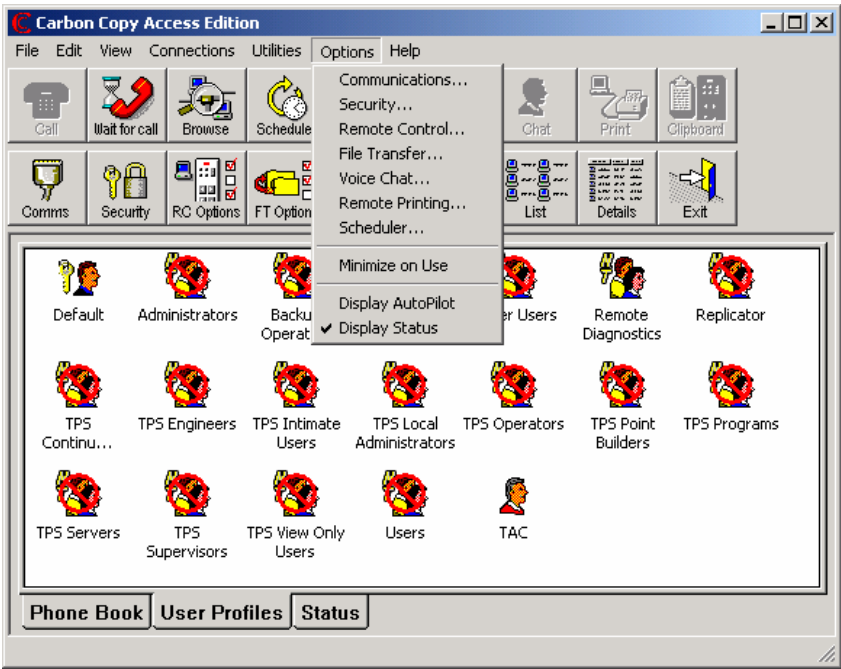
ATTENTION:

You may encounter a Security dialog box that reads like this: "You have set Carbon Copy to use native NT security. Carbon Copy needs to run as a service under this configuration. Please exit the program and restart it as a service. If so, Start>Programs>Carbon Copy Access Edition>Start Service."

If you encounter this message, click **OK** and restart your computer. After logging on to the machine, continue to the next step.

Step	Action
Configure Communication Properties	

4 Select **Options>Communications**.



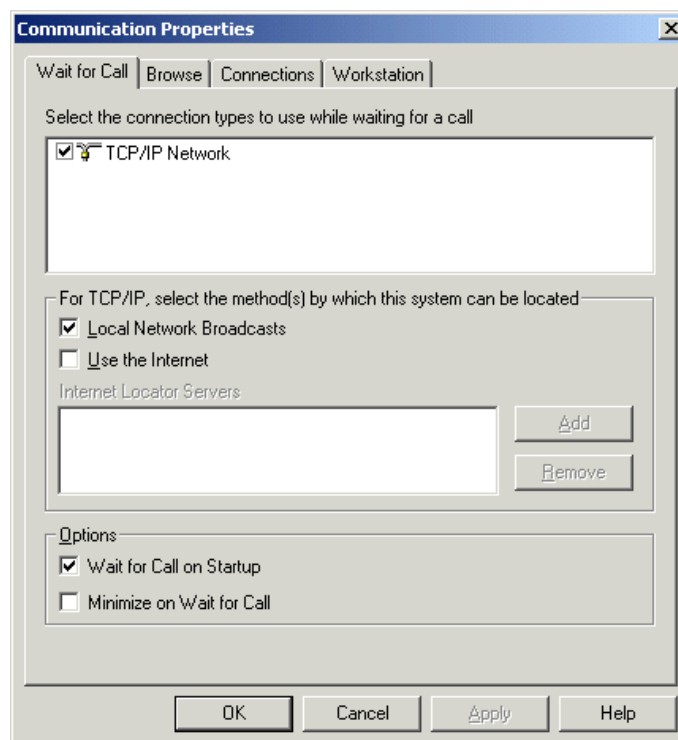
2 Instructions for Initial Remote GUS Setup

2.6 Configure Carbon Copy AE

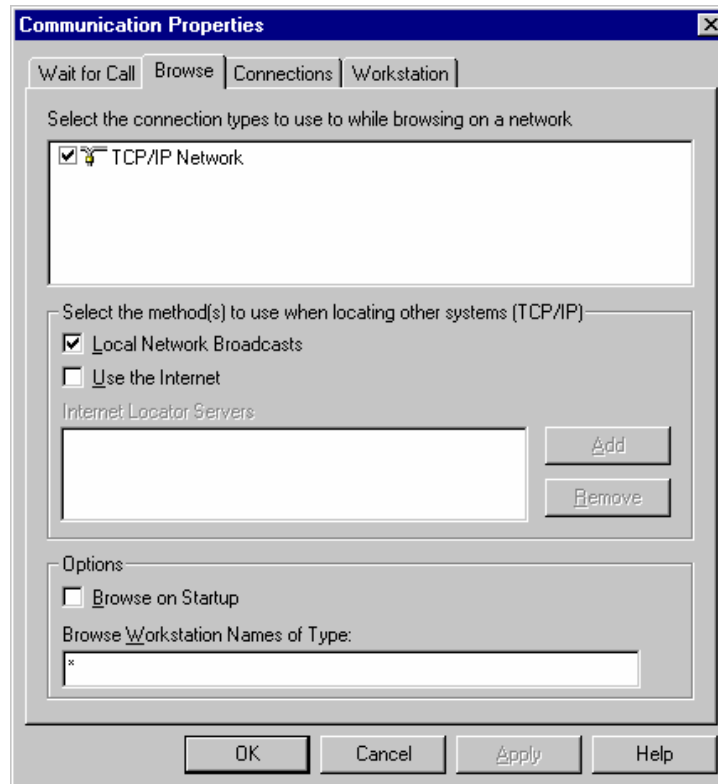
Step	Action
------	--------

- | | |
|---|---|
| 5 | <p>Click the Wait for Call tab.</p> <ul style="list-style-type: none">• Check all available checkboxes <i>except</i> Use the Internet and Minimize on wait for Call.• If there are any ILS links in the Internet Locator Servers list box, select them and Remove them. |
|---|---|

Click the **Apply** button, if available.



- | Step | Action |
|------|--|
| 6 | <p>Click the Browse tab and view the options.</p> <ul style="list-style-type: none">• Check all available checkboxes, <i>except</i> Browse on Startup.• If there are any ILS links in the Internet Locator Servers list box, select them and Remove them.• If there are any entries (except an initial asterisk *) in the Browse Workstation Names of Type list box, delete them. <p>Click the Apply button, if available.</p> |



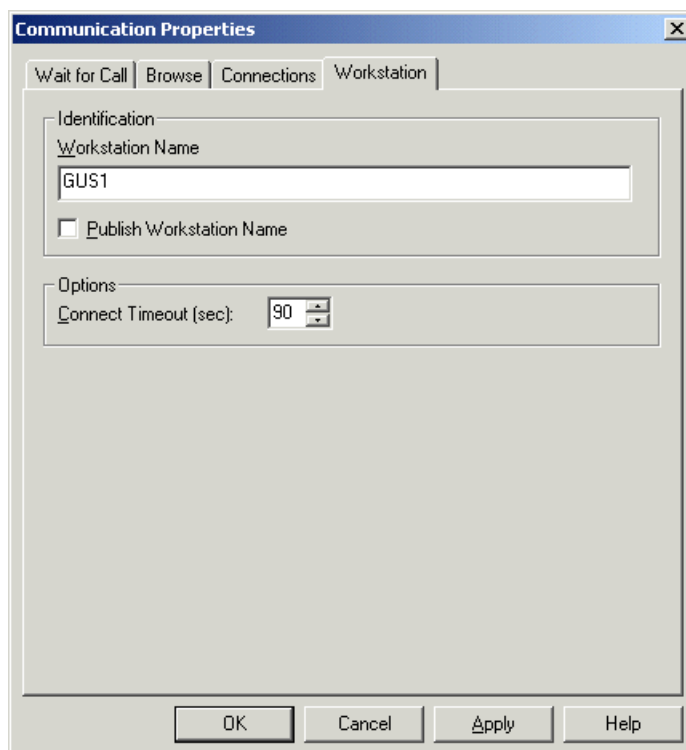
- 7 The **Connections** tab needs no configuration.

2 Instructions for Initial Remote GUS Setup

2.6 Configure Carbon Copy AE

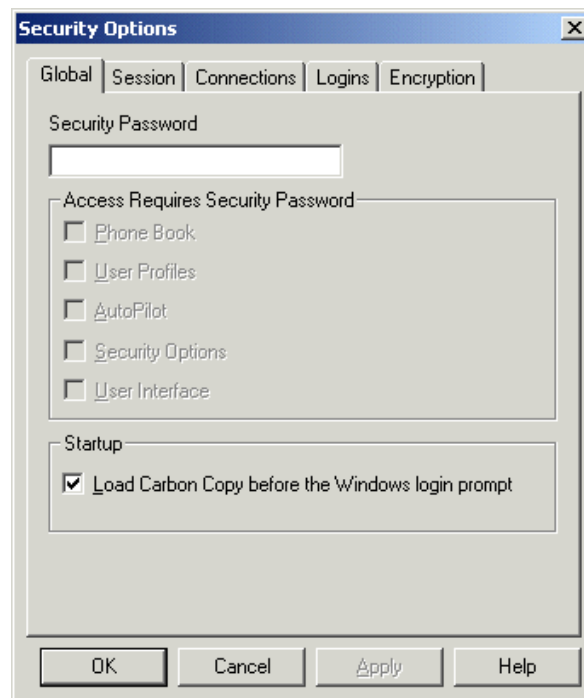
Step	Action
------	--------

- | | |
|---|--|
| 8 | <p>Click the Workstation tab.</p> <ul style="list-style-type: none">• The local workstation name should be in the Workstation Name port; if not, add it.• Publish Workstation Name may be checked or unchecked.• Set Connect Timeout to 90. |
|---|--|



- | | |
|---|---|
| 9 | <p>Click Apply. Click the OK button to close the Communication Properties dialog box.</p> |
|---|---|

Step	Action
Configure Security Options	
10	From the menu bar, select Options>Security .
11	Click the Global tab. • Leave the Security Password blank. • Check the checkbox for Load Carbon Copy before the Windows login prompt.

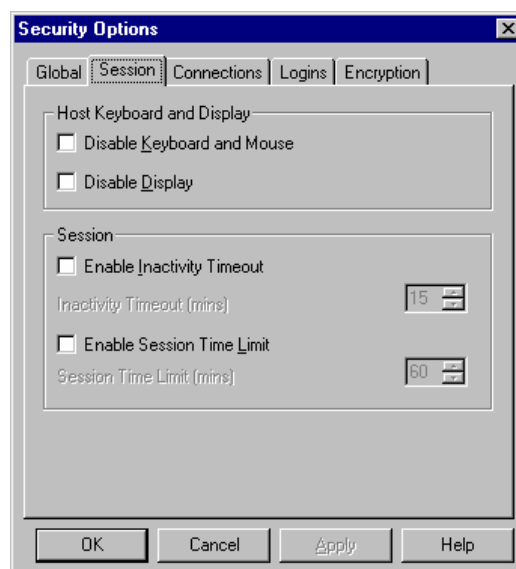


2 Instructions for Initial Remote GUS Setup

2.6 Configure Carbon Copy AE

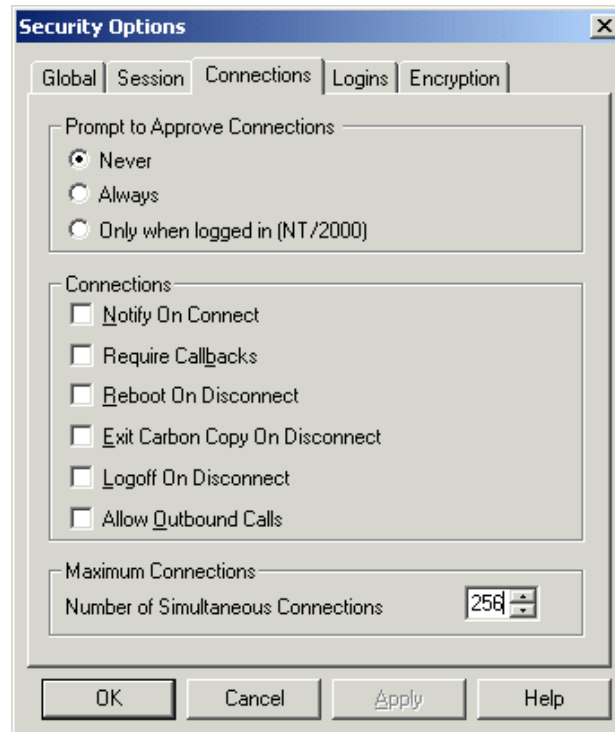
Step	Action
------	--------

- | | |
|----|--|
| 12 | <p>Click the Session tab.</p> <p>Uncheck (or leave unchecked) all possible entries.</p> <p>Click Apply.</p> |
|----|--|



Step	Action
------	--------

- | | |
|----|---|
| 13 | <p>Click the Connections tab.</p> <ul style="list-style-type: none">• Select the Never radio button under Prompt to Approve Connections.• Leave all boxes unchecked under Connections, except the following:<ul style="list-style-type: none">– <i>On Client machine</i>, check Allow Outbound Calls.• Set the Number of Simultaneous Connections to 256. <p>Click Apply.</p> |
|----|---|



2 Instructions for Initial Remote GUS Setup

2.6 Configure Carbon Copy AE

Step	Action
------	--------

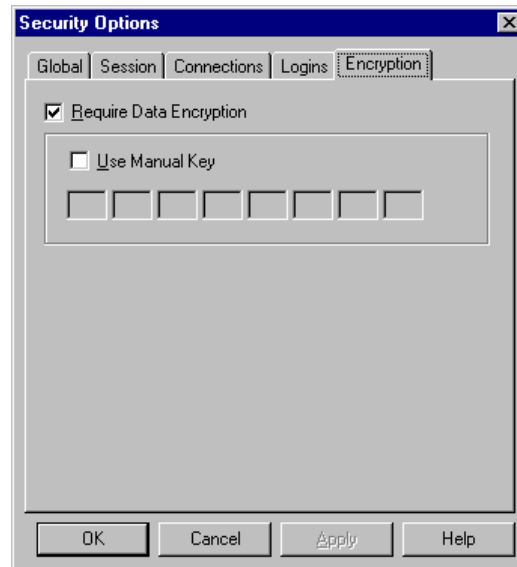
- | | |
|----|--|
| 14 | <p>Click the Logins tab.</p> <ul style="list-style-type: none">• Check Require Logins.• Select the Use Native NT/2000 security.• Uncheck (leave unchecked) Logoff NT on Connect.• Set Login Attempts Allowed to 3.• Set Login Timeout (secs) to 30.• Accept default in Minimum password length required (may be grayed out).• Uncheck (leave unchecked) Enable Lockout Security. |
|----|--|



Step	Action
------	--------

15 Click the **Encryption** tab and set or clear the following checkmarks.

- Check **Require Data Encryption**.
- Uncheck (leave unchecked) **Use Manual Key**.



16 Click **Apply**. Click the **OK** button to close the Security Options dialog box.

Configure Remote Control

17 From the menu bar, select **Options>Remote Control**.

18 Click the **Desktop and Cache** tab and set or clear the following:

Check:

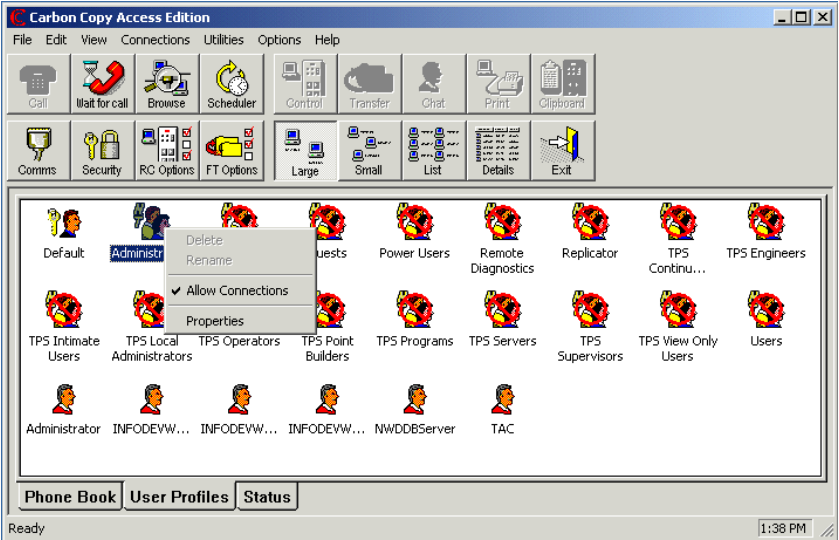
- **Disable Wallpaper on Host**
- **Disable screen saver on Host**
- **Disable Active Desktop on Host**
- **Disable Font Smoothing on Host**
- **Ignore Large Bit maps**

Set **Smallest Bitmap to Ignore** (KPixels) to **64**

2 Instructions for Initial Remote GUS Setup

2.6 Configure Carbon Copy AE

Step	Action
	Uncheck (or leave unchecked): • Track Blinking Cursors • Show Host Mouse Motion. • Synchronize Host and Guest Displays. • Use Accelerated Fonts. Check Enable Disk Cache Set Size of Disk Cache (Mbytes) to 10.
19	The other tabs do not require configuration: Screen, Hot Keys, DOS, and Host.
20	Click the OK button to close the dialog box.
	Enable Account
21	Click the User Profiles tab at the bottom of the Carbon Copy window. RESULT: The User Profiles page shows all the Groups and Users configured on this machine.
22	Select the Administrators group.

Step	Action
23	Right-click on the Administrators icon and select Allow Connections. ATTENTION: The red circle with the slash through it should disappear on the Administrators icon and whatever users assigned to the Administrators Group should appear.  The screenshot shows the 'Carbon Copy Access Edition' application window. The menu bar includes File, Edit, View, Connections, Utilities, Options, and Help. Below the menu bar is a toolbar with icons for Call, Wait for call, Browse, Scheduler, Control, Transfer, Chat, Print, and Clipboard. A second row of icons includes Comms, Security, RC Options, FT Options, Large, Small, List, Details, and Exit. The main area displays a grid of user and group icons. The 'Administrators' icon is highlighted, and a right-click context menu is open over it, showing options: Delete, Rename, Allow Connections (checked), and Properties. Other icons visible include Default, Guests, Power Users, Remote Diagnostics, Replicator, TPS Continu..., TPS Engineers, TPS Intimate Users, TPS Local Administrators, TPS Operators, TPS Point Builders, TPS Programs, TPS Servers, TPS Supervisors, TPS View Only Users, Users, Administrator, INFODEVW..., INFODEVW..., INFODEVW..., NWDBServer, and TAC. At the bottom are buttons for Phone Book, User Profiles, and Status. The status bar at the bottom left says 'Ready' and the bottom right shows the time '1:38 PM'.
24	Select the Remote Diagnostics group.
25	Right-click on the Remote Diagnostics icon and select Allow Connections. ATTENTION: The red circle with the slash through it should disappear on the Remote Diagnostics icon and whatever users assigned to the Remote Diagnostics Group should appear.
26	Continue to section 3.1 Introduction to Remote GUS Security.

2 Instructions for Initial Remote GUS Setup

2.6 Configure Carbon Copy AE

3. Remote GUS Security Configuration

3.1 Introduction to Remote GUS Security

Accounts and groups

The Carbon Copy AE Host (GUS Server) uses Windows security to authenticate a login from a Carbon Copy AE guest (GUS Client).

You may create GUS Server (Host) user accounts or use one of the standard TPS user accounts (such as the GUS local account or an Operator or Engineer account).

The GUS Server (Host) user account must be a member of a GUS Server (Host) group. During the setup of Carbon Copy AE GUS Server (Host) components, you configure the group to “Allow Connections.”

The TAC user account is used to log on to a manned GUS Server (Host) from a remote location for use in remote engineering and technical assistance scenarios.

The TAC user account allows the assignment of user rights within Carbon Copy AE for tasks such as file transfer.



ATTENTION

Honeywell recommends using a password with every GUS Server (Host) user account that is created.

GUS Client Security Configuration

Only users that belong to the Windows Power User or Administrators group may start/stop and select IKB/OEP connection to the host. For details, refer the section 5.3 Configure Remote IKB Client.

3 Remote GUS Security Configuration

3.1 Introduction to Remote GUS Security

Honeywell recommends the following Windows login security configuration.

Security Area	Recommendation
Windows Login Security	Define operator access rights at the GUS Client according to local policy (as described in this section). Operator user account can be a TPS Domain account, if GUS Client has been configured to be a member of the TPS domain.

GUS Server Security Configuration

Perform one of the following procedures to configure the GUS Server local policy to allow the Remote Diagnostics group to have the user right to log on locally.

- Section 3.2 - Configure GUS Server Local Policy (Using Local Security Policy)
- Section 3.3 - Configure GUS Server Local Policy (using Security Configuration & Analysis)
- Section 3.4 - Configure GUS Server Local Policy (Using Local Security Policy – with Administrators set to DENY ACCESS)



ATTENTION

The procedures in Section 3 assume the following:

- A Honeywell TAC user is a member of the Remote Diagnostics group.
- Carbon Copy AE is setup to allow connections for the Remote Diagnostics group (**User Profiles** tab, right-click on the **Remote Diagnostics** group, and then check **Allow Connections**).

You can create other groups (for example, Remote Maintenance, Remote Engineers, and Remote Operators), and then give those groups the “rights” you want them to have.

3.2 Configure GUS Server Local Policy (Using Local Security Policy)

Purpose

The following procedure configures a policy to allow the members of the Remote Diagnostics group to log on locally to the GUS Server.

Procedure

Step	Action
1	At the GUS Server machine, display the Local Security Settings MMC by selecting Start>Programs>Administrative Tools>Local Security Policy .
2	In the Console Tree of the Local Security Settings MMC, expand Local Policies , and then click on User Rights Assignment .
3	In the Details Pane of the Local Security Settings MMC, double-click Log on locally .
4	In the Local Security Policy Setting dialog box, click the Add button.
5	In the Select Users or Groups dialog box, make sure the 'Look in:' box indicates your GUS. Locate and select the Remote Diagnostics group. Click the Add button and then click the OK button.
6	In the Local Security Policy Setting dialog box, make sure the checkbox for the GUS\Remote Diagnostics group (in the Local Policy Setting column) is selected.

ATTENTION:

Notice that the checkbox in the Effective Policy Setting column is not selected. You need to refresh the computer policy to make the new setting effective.

Click the **OK** button.

Refresh the Computer Policy

- | | |
|---|--|
| 7 | Close the Local Security Settings MMC. |
| 8 | Select Start>Run . |
| 9 | In the Run dialog box, enter Secedit /refreshpolicy machine_policy , and click the OK button. |

3 Remote GUS Security Configuration

3.2 Configure GUS Server Local Policy (Using Local Security Policy)

Step	Action
Verify New Setting	
10	Display the Local Security Settings MMC by selecting Start>Programs>Administrative Tools>Local Security Policy .
11	In the Console Tree of the Local Security Settings MMC, expand Local Policies , and then click on User Rights Assignment .
12	In the Details Pane of the Local Security Settings MMC, double-click Log on locally .
13	In the Local Security Policy Setting dialog box, make sure the checkbox for the GUS\Remote Diagnostics group (in the Effective Setting column) is selected and then click the Cancel button.
14	Close the Local Security Settings MMC
15	Continue to the section 4.1 Create a New Connection.

3.3 Configure GUS Server Local Policy (using Security Configuration & Analysis)

Purpose

The following procedure configures a policy to allow the members of the Remote Diagnostics group to log on locally to the GUS Server.

Procedure

Step	Action
Open Security Configuration and Analysis MMC	
1	At the GUS Server machine, select Start>Run .
2	In the Run dialog box, type mmc and press Enter .
3	In the Console1 window, select Console>Add/Remove Snap-in .
4	In the Add/Remove Snap-in dialog box, click the Add button.
5	In the Add Standalone Snap-in dialog box, find and select Security Configuration and Analysis and click the Add button. Click the Close button.
6	In the Add/Remove Snap-in dialog box, click the OK button.
7	Maximize the Console Root window.
Open TPSSDB database in Security Configuration and Analysis MMC	
8	Right-click the Security Configuration and Analysis scope item and select Open database .
9	In the Open database dialog box, select C:\winnt\security\database\tpssdb.sdb and click the Open button.
Compare Database Settings to Computer Settings	
10	Right-click the Security Configuration and Analysis scope item and select Analyze Computer Now .
11	In the Perform Analysis, accept the default path and click the OK button.
12	In the Console1 window, expand Account Policies .
Find Differences between Database Settings and Computer Settings	

3 Remote GUS Security Configuration

3.3 Configure GUS Server Local Policy (using Security Configuration & Analysis)

Step	Action
13	Click on the Password Policy scope item and look for policies whose Database Setting and Computer Setting do not agree. ATTENTION: A "green check" appears on policies whose Database Setting and Computer Setting agree. A "red X" (actually a red Circle with a white X) appears on policies whose Database Setting and Computer Setting do NOT agree.
14	Continue expanding scope items looking for policies with a "red X".
15	For each policy with a "red X", follow the steps in the next task to make the Database Settings to match the Computer Settings.
Change Database Settings to match Computer Settings (For each policy with a "red X")	
16	Double click on the policy (which has a "red X").
17	Make the Database Setting match the Computer Setting and click the OK button.
Check the Configuration.	
18	Analyze the computer and look for any differences between the database and the computer settings. Continue to the next step only when no differences are found.
Assign "Log on locally" user right to Remote Diagnostics group (in the Database)	
19	Expand Local Policies>User Rights Assignment .
20	Double click on the Log on locally policy.
21	In the Analysis Security Policy Setting dialog box, click the Add button.
22	In the Select Users or Groups dialog box, select your GUS from the 'Look in:' pull down list.
23	Locate and select the Remote Diagnostics group, click the Add button, then click the OK button.
24	In the Analysis Security Policy Setting dialog box, click the OK button.
Configure computer to assign "Log on locally" user right to Remote Diagnostics group (in the computer)	

3 Remote GUS Security Configuration

3.3 Configure GUS Server Local Policy (using Security Configuration & Analysis)

Step	Action
25	In the Console1 window, right-click the Security Configuration and Analysis scope item and select Configure Computer Now .
26	Analyze the computer and verify that the Remote Diagnostics group has been assigned the "Log on locally" user right.
27	Close the Console1 window. Do not save console settings.
28	Continue to the section 4.1 Create a New Connection.

3 Remote GUS Security Configuration

3.4 Configure GUS Server Local Policy (Using Local Security Policy – with Administrators set to DENY ACCESS)

3.4 Configure GUS Server Local Policy (Using Local Security Policy – with Administrators set to DENY ACCESS)

Purpose

The following procedure configures a policy to allow the members of the Remote Diagnostics group to log on locally to the GUS Server.

Procedure

Step	Action
1	Using Windows Explorer, navigate to the following folder: C:\WINNT\system32\GroupPolicy ATTENTION: You will not be able to open the folder.
2	Display the properties of the GroupPolicy folder.
3	Use the Apply button (NOT the OK button) to modify the permissions of the GroupPolicy folder to allow the Administrators group access to the folder (remove the Deny access from all checkboxes).  WARNING: Leave the GroupPolicy properties open after applying the changes. If you do not, it is possible that the very restrictive local policy could be applied to the administrators group.
4	Display the Local Security Settings MMC by selecting: Start>Programs>Administrative Tools>Local Security Policy.
5	In the Console Tree of the Local Security Settings MMC, expand Local Policies , and then click on User Rights Assignment .
6	In the Details Pane of the Local Security Settings MMC, double-click on Log on locally .
7	In the Local Security Policy Setting dialog box, click on the Add button.
8	In the Select Users or Groups dialog box, make sure the 'Look in:' box indicates the GUS. Locate and select the Remote Diagnostics group, click the Add button, and then click the OK button.

3 Remote GUS Security Configuration

3.4 Configure GUS Server Local Policy (Using Local Security Policy – with Administrators set to DENY ACCESS)

Step	Action
9	In the Local Security Policy Setting dialog box, make sure the checkbox for the GUS\Remote Diagnostics group (in the Local Policy Setting column) is selected and click the OK button. ATTENTION: Notice that the checkbox in the Effective Policy Setting column is not selected. You will need to refresh the computer policy to make our new setting effective.
10	Close the Local Security Settings MMC.
Refresh the Computer Policy	
11	Select Start>Run .
12	In the Run dialog box, enter Secedit /refreshpolicy machine_policy , and click the OK button.
Verify New Setting	
13	Display the Local Security Settings MMC by selecting: Start>Programs>Administrative Tools>Local Security Policy
14	In the Console Tree of the Local Security Settings MMC, expand Local Policies , and then click on User Rights Assignment .
15	In the Details Pane of the Local Security Settings MMC, double-click Log on locally .
16	In the Local Security Policy Setting dialog box, make sure the checkbox for the GUS\Remote Diagnostics group (in the Effective Setting column) is selected and click the Cancel button.
17	Close the Local Security Settings MMC.
18	In the GroupPolicy properties dialog box, deny full control to the administrators group.
	WARNING: The above step is very important. If you do not do this step, you will apply the very restrictive local policy to the administrators group.
19	Close the GroupPolicy properties and Windows Explorer.
20	Continue to the section 4.1 Create a New Connection.

3.5 Security Options for Manned or Unmanned Operation

Purpose

Options on the Carbon Copy security options Connections tab are used by the "Remote Control" utility after connecting to a remote PC.

You may set the host security options for Manned Host or Unmanned Host operations:

Manned host

With manned host configuration, the local host user must explicitly acknowledge each guest connection request. Example: For connection to Honeywell TAC assistance.

Unmanned host

With unmanned host configuration, the local host user acknowledgement is not required. Example: For remote operations.

Manned or Unmanned Host

Carbon Copy AE GUS Client host components are configured for manned operation.

Carbon Copy AE GUS Server host components can be configured for manned or unmanned operation:

- host components are configured for *manned* operation for remote engineering and remote TAC assistance applications.
- host components are configured for *unmanned* operation for remote operations applications.

Procedure to set Security Options

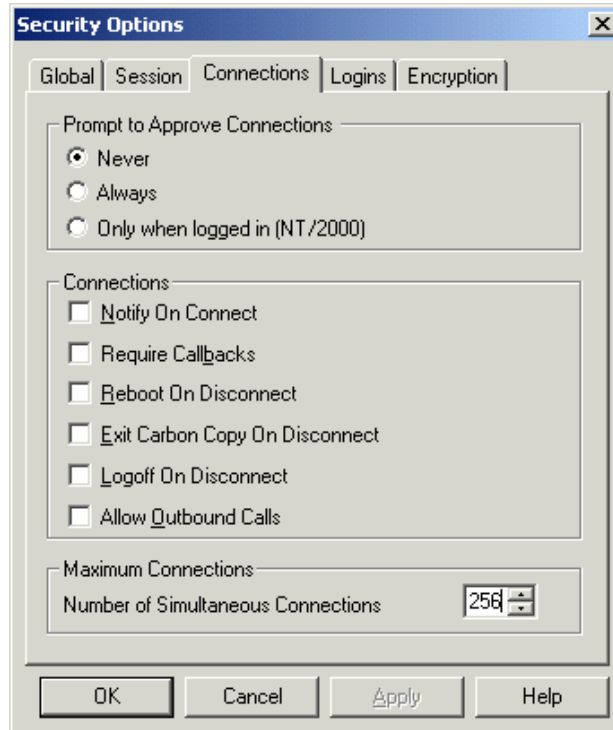
Step	Action
1	Call up the Carbon Copy AE user interface (launch the program) by selecting: Start>Programs>Carbon Copy Access Edition>Carbon Copy
2	From the Carbon Copy AE menu bar, select Options>Security

3 Remote GUS Security Configuration

3.5 Security Options for Manned or Unmanned Operation

Step	Action
------	--------

3	Select the Connections tab.
---	------------------------------------



- | | |
|---|---|
| 4 | <p>Set the options on a manned host as follows:</p> <ul style="list-style-type: none">• Prompt to Approve Connections – Set to Always.• Number of Simultaneous Connections – Set to 1.• Uncheck all of the other options. <p>Set the options on a unmanned host as follows:</p> <ul style="list-style-type: none">• Logoff On Disconnect• Number of Simultaneous Connections – Set to 1.• Uncheck all of the other options. |
|---|---|
-

3 Remote GUS Security Configuration

3.6 Limit Access to Carbon Copy Installation Folder

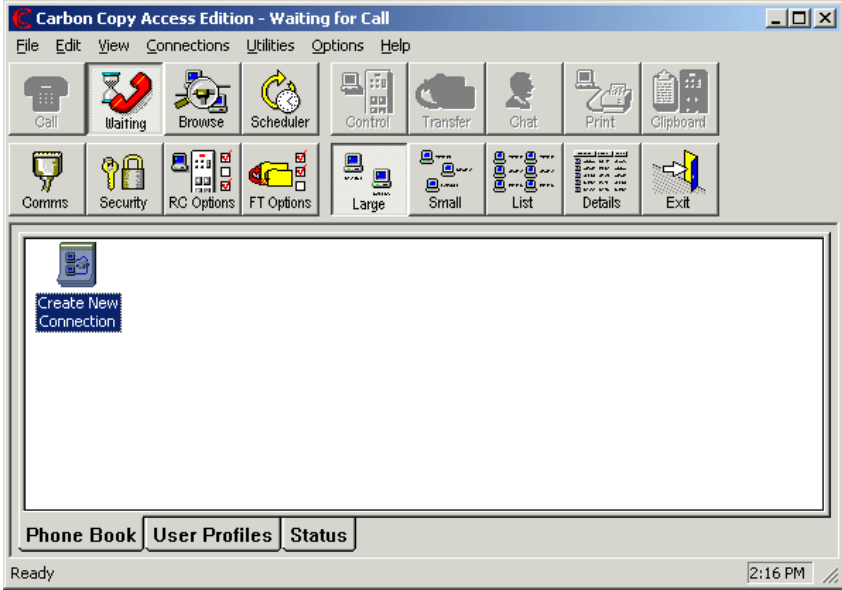
3.6 Limit Access to Carbon Copy Installation Folder

It is recommended that the following procedure be used to limit access to the Carbon Copy AE installation folder.

Step	Action
1	Use Windows Explorer to navigate to the following Carbon Copy installation folder: c:\program files\carbon copy
2	Right-click on the folder.
3	Select Properties .
4	Select the Security tab.
5	Verify that the permissions are as follows: Everyone: Read Administrators: Full Control

4. Setting Up a Connection

4.1 Create a New Connection

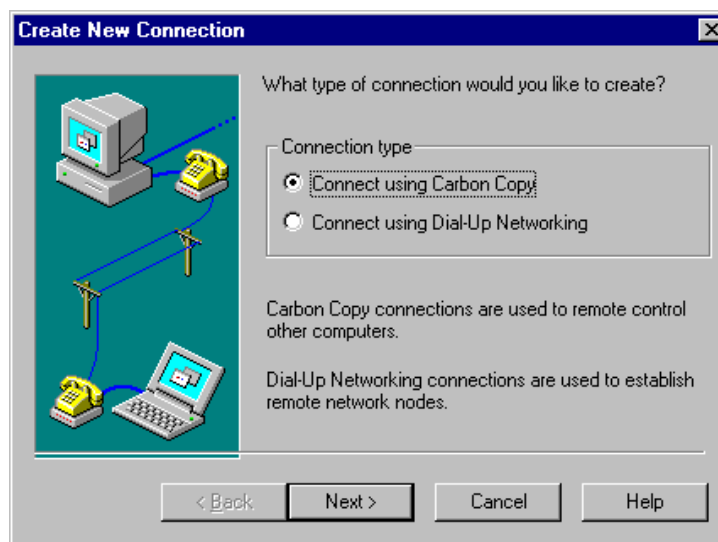
Step	Action
1	Log in at the GUS Client as the Local Administrator.
2	If the Tray icon  1:02 PM is present in the lower right corner, launch Carbon Copy AE from it. If not present, select: Start>Programs>Carbon Copy Access Edition>Carbon Copy RESULT: The Carbon Copy control window opens.
	
3	Select the Phone Book tab.
4	Double-click the Create New Connection icon.
5	<i>The remaining steps differ, depending on the type of the connection: LAN, RAS, or modem.</i> Continue to the section 4.2 Create a New TCP/IP LAN Connection.

4 Setting Up a Connection

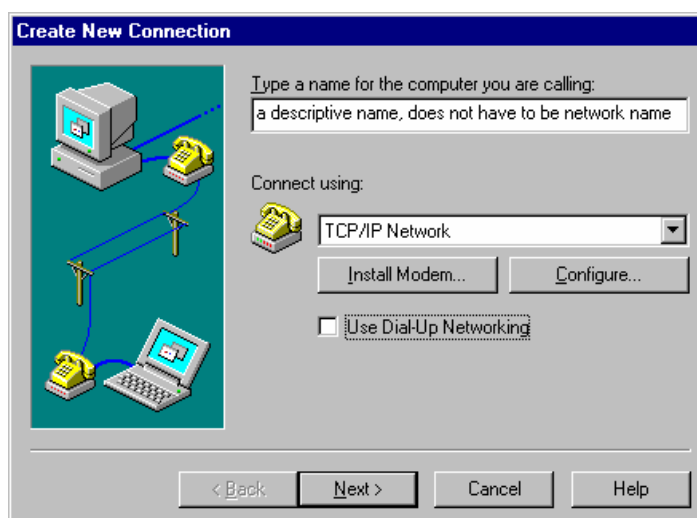
4.2 Create a New TCP/IP LAN Connection

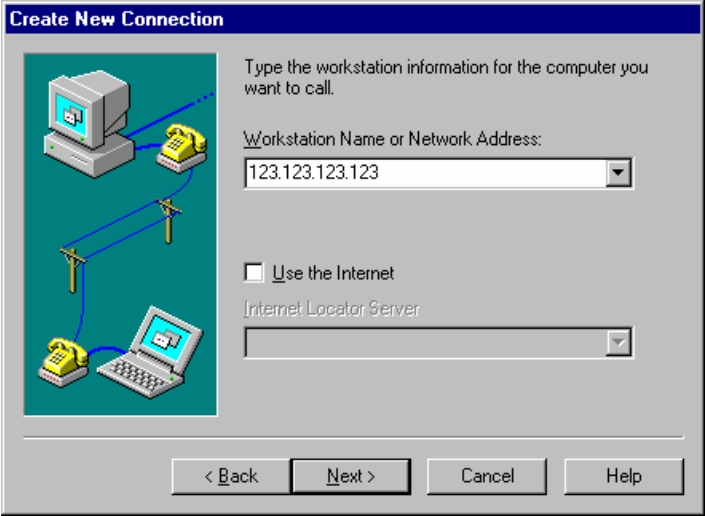
4.2 Create a New TCP/IP LAN Connection

- | Step | Action |
|------|--|
| 1 | Select the following: Connect using Carbon Copy |



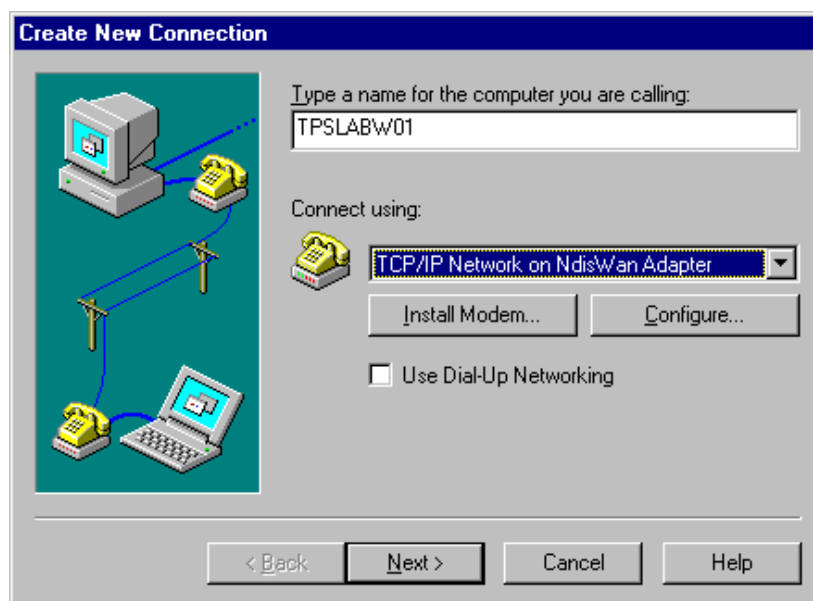
- 2 If connecting with a TCP/IP LAN connection, you will see the following.



Step	Action
	ATTENTION: - In the above example, the Carbon Copy AE client resides on a TCP/IP network that has RAS and a US Robotics 56K external FAX modem installed. - The LAN network, RAS network, and modem must have been installed earlier. - Reinstall TCP/IP if TCP/IP does not appear.
3	Type a descriptive name for the connection, such as the name of the host. (The name will be used to label the icon on the Phone Book page.)
4	Click Next .
5	Enter the Workstation Name or TCP/IP address of the Gus Server (Host).
	
6	Click the Next button.
7	Continue to the next section 4.3 Create a New RAS Connection.

4.3 Create a New RAS Connection

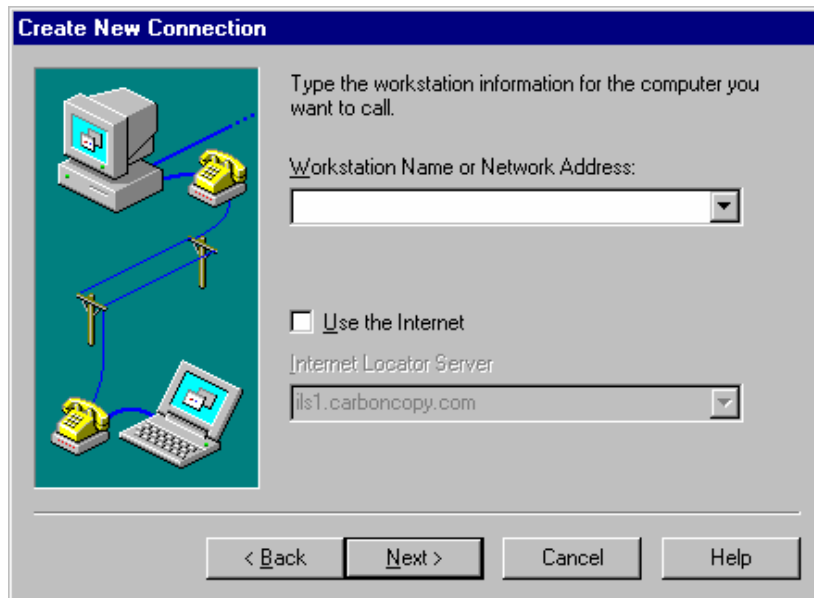
- | Step | Action |
|------|---|
| 1 | Double-click the Create New Connection icon. |
| 2 | Type a name for your connection. |
| 3 | Select TCP/IP Network on NdisWan Adapter . |



- | | |
|---|-------------------------------|
| 4 | Click the Next button. |
|---|-------------------------------|

Step	Action
5	Enter the GUS Server (Host) RAS IP address.

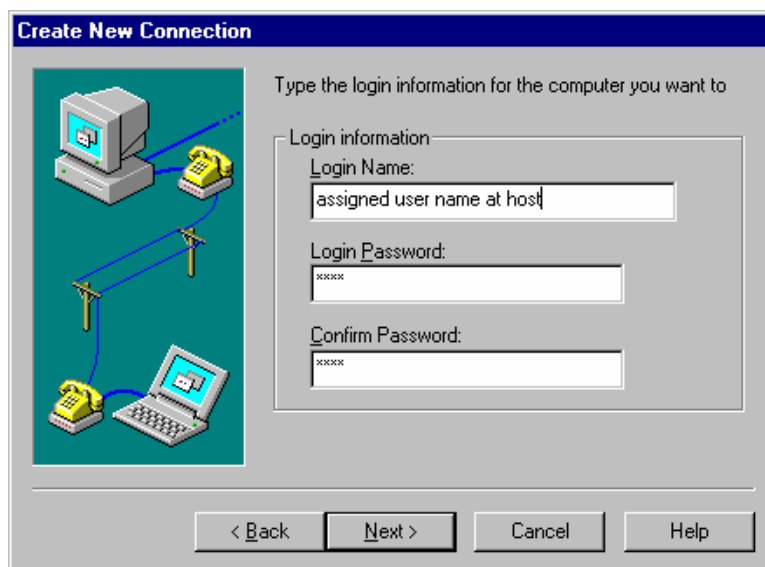
ATTENTION: If you leave it blank, you will always be prompted for the Workstation Name or Network Address when you connect.



- 6 Click the **Next** button.
- 7 Continue to the section 4.4 Complete the New Connection.

4.4 Complete the New Connection

- | Step | Action |
|------|---|
| 1 | Enter the Login Name and Login Password of the account you set up in Carbon Copy USER PROFILES on the GUS Server (Host). |



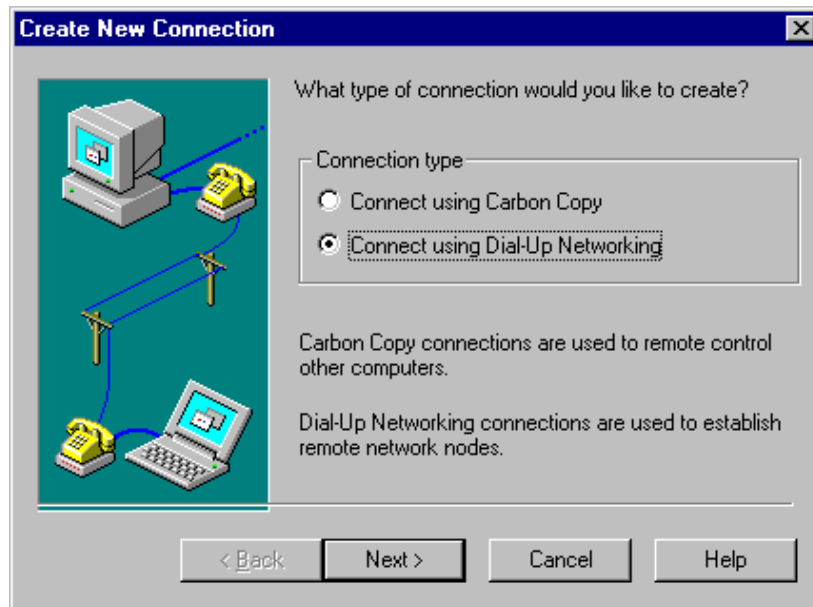
- | | |
|---|--|
| 2 | Click the Next button. |
| 3 | Check the box for Remote Control .

(By doing this, when Carbon Copy starts, you will see a window with only the host's computer screen on it.) |
| 4 | Click the Next button. |
| 5 | Click the Finish button to complete the connection creation. |
| 6 | Continue to the section 4.5 Create a New Modem Connection. |
-

4.5 Create a New Modem Connection

Step	Action
------	--------

- | | |
|---|---|
| 1 | Double-click the Create New Connection icon to display the following. |
|---|---|



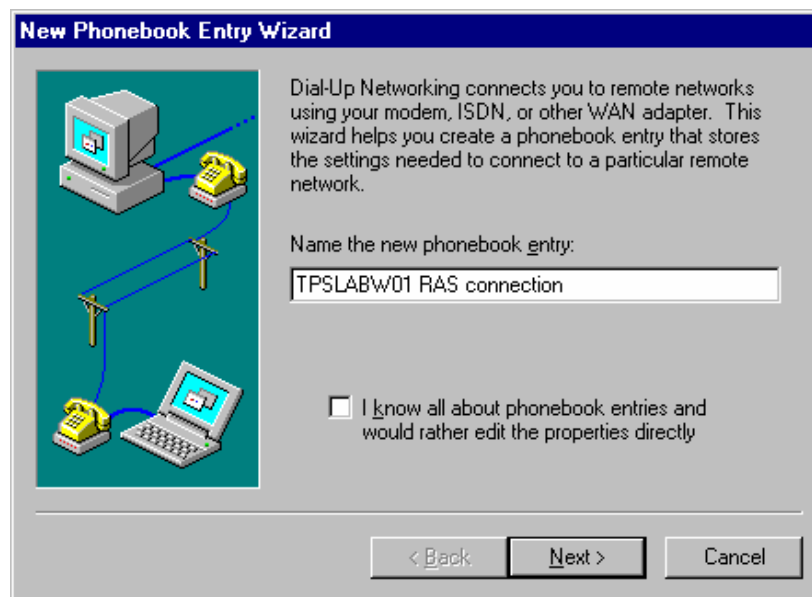
- | | |
|---|--|
| 2 | Select the following: Connect using Dial-Up Networking
ATTENTION: This is needed to first establish the RAS link. |
| 3 | Click Next . |

4 Setting Up a Connection

4.5 Create a New Modem Connection

Step	Action
------	--------

- | | |
|---|--|
| 4 | If you will connect with a TCP/IP LAN connection, then you will see the following: |
|---|--|



- | | |
|---|---|
| 5 | Enter a descriptive name, such as the name of the host. |
| 6 | Click the Next button. |

- | Step | Action |
|------|--|
| 7 | Click the Next button again to display the following: |

Phone Number

Enter the phone number of the dial-up server you are calling. Alternate phone numbers, if any, are dialed automatically if the primary phone number cannot be reached. They may also be used to set different numbers on individual multi-linked ISDN lines.

Country code:
United States of America (1)

Area code: 612 Phone number: 555 2323

Alternates...

☒ Use Telephony dialing properties

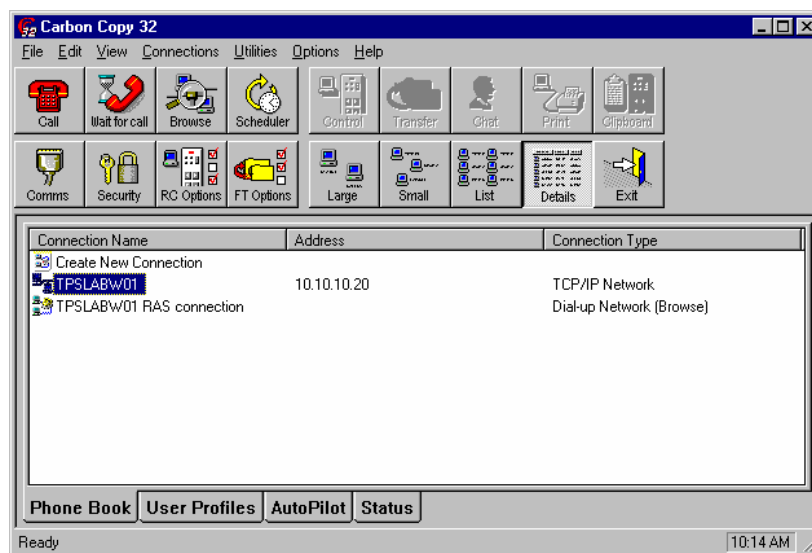
< Back Next > Cancel

- | | |
|----|--|
| 8 | Uncheck Use Telephony dialing properties . |
| 9 | Enter the Phone number . |
| 10 | Click the Next button. |
| 11 | Click the Finish button. |
| 12 | Continue to the section 4.6 Configure Connection Properties. |
-

4.6 Configure Connection Properties

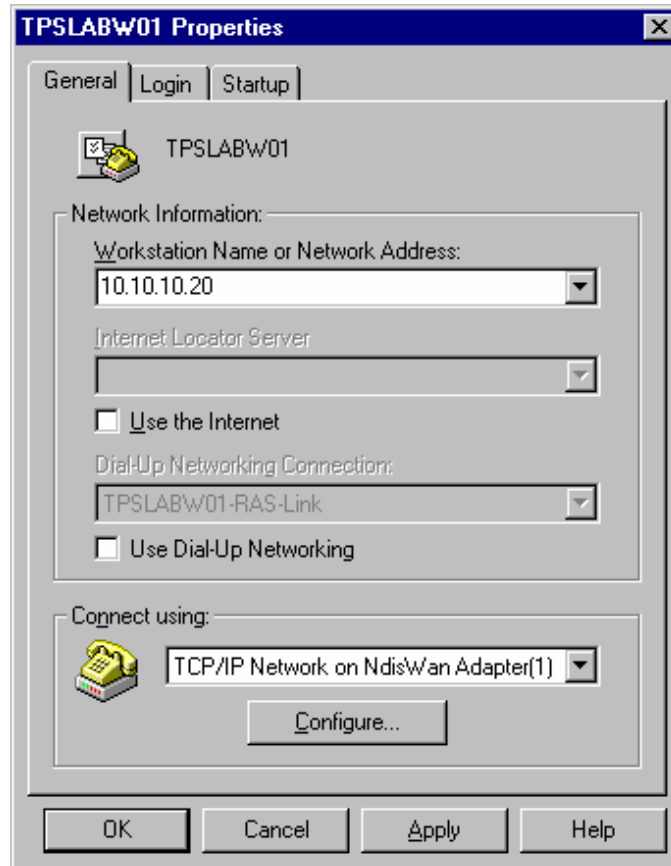
Step	Action
------	--------

- | | |
|---|---|
| 1 | You should see two links: One that resembles a LAN connection and one that resembles a dialup modem connection. |
|---|---|



- | | |
|---|---------------------------------|
| 2 | Select the LAN connection name. |
|---|---------------------------------|

- | Step | Action |
|------|--|
| 3 | Right-click and select Properties to display the following: |

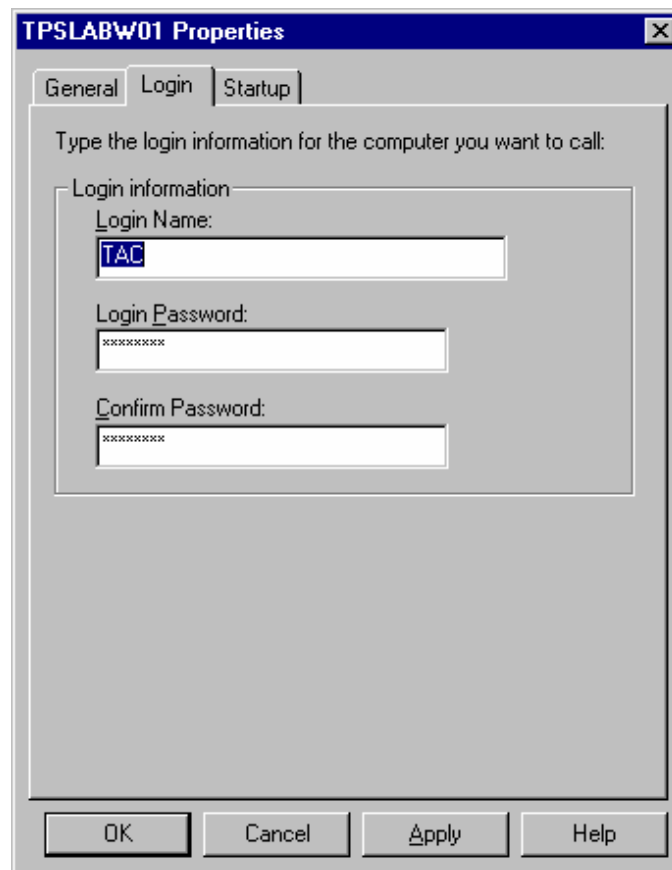


- | | |
|---|---|
| 4 | Verify that under Connect using , TCP/IP Network on NdisWan Adapter is selected. |
|---|---|

4 Setting Up a Connection

4.6 Configure Connection Properties

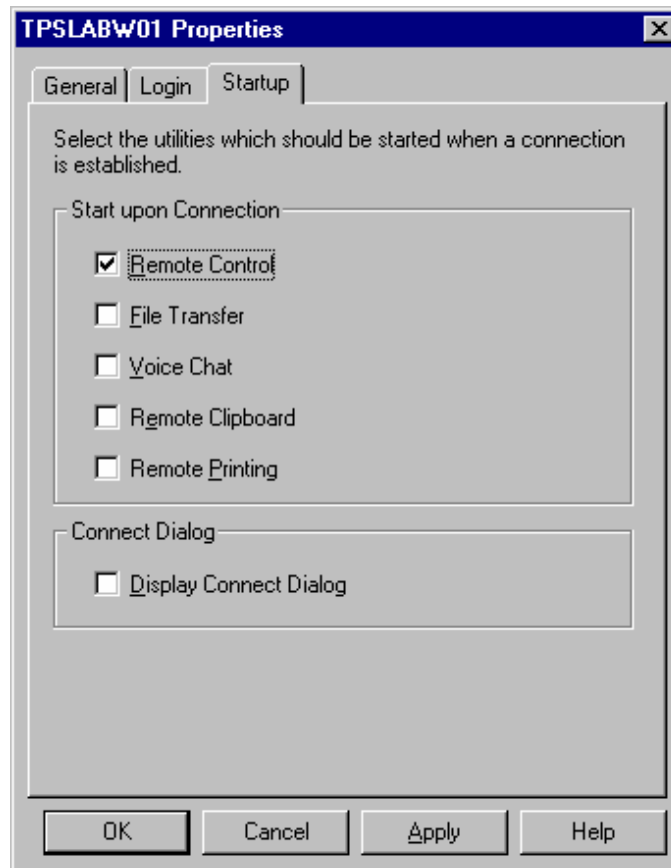
- | Step | Action |
|------|---|
| 5 | Select the Login tab to display the following: |



The image shows a Windows-style dialog box titled "TPSLABW01 Properties". It has three tabs: "General", "Login", and "Startup". The "Login" tab is selected. Inside the dialog, there is a text prompt: "Type the login information for the computer you want to call:". Below this, there is a section titled "Login information:" which contains three input fields. The first field is labeled "Login Name:" and contains the text "TAC". The second field is labeled "Login Password:" and contains a series of asterisks "xxxxxxxx". The third field is labeled "Confirm Password:" and also contains a series of asterisks "xxxxxxxx". At the bottom of the dialog, there are four buttons: "OK", "Cancel", "Apply", and "Help".

- 6 If not previously entered, enter the **Login Name**.
- ATTENTION:** The Login Name may require domain\username.
- 7 If not previously entered, enter the **Login Password**.

- | Step | Action |
|------|--|
| 8 | Select the Startup tab to display the following: |

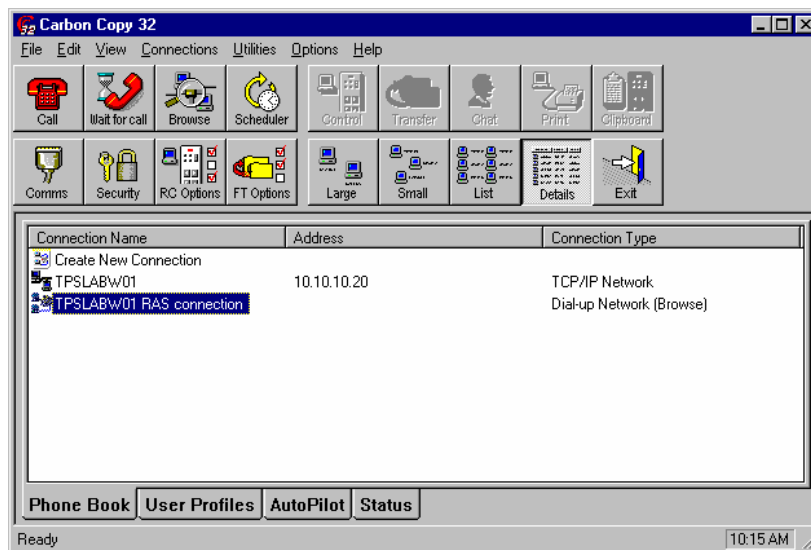


- 9 Uncheck the following:
- Display Connect Dialog**

4 Setting Up a Connection

4.6 Configure Connection Properties

- | Step | Action |
|------|--|
| 10 | Click the OK Dialog to display the following: |



Configure Dialup Modem Connection Properties

- 11 Select the Dialup-modem connection name.
- 12 Right-click and select **Properties**.

- | Step | Action |
|-----------|---|
| 13 | Select the Basic tab to display the following: |

Edit Phonebook Entry

Basic | Server | Script | Security | X.25

Entry name: TPSLABW01 RAS connection

Comment:

Country code: United States of America (1)

Area code: 612

Phone number: 555 2323 Alternates...

☒ Use Telephony dialing properties

Dial using: Sportster 56000 Fax External (COM) Configure...

☐ Use another port if busy

OK Cancel

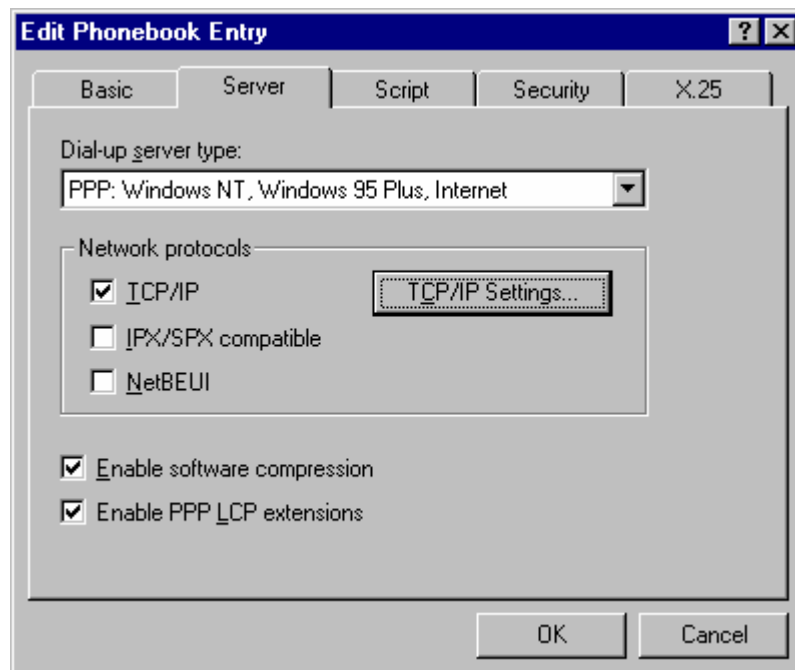
- 14** Uncheck the following:
- Use another port if busy**

4 Setting Up a Connection

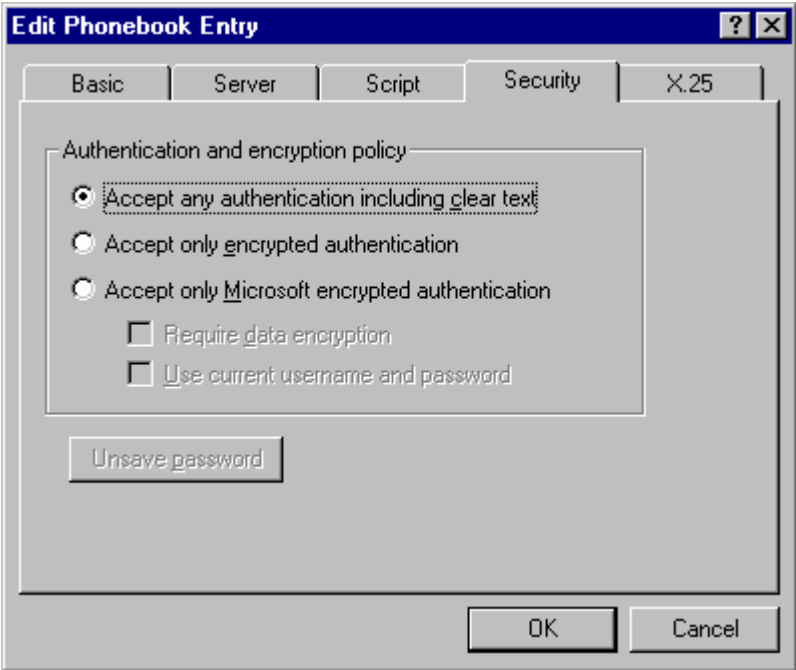
4.6 Configure Connection Properties

Step	Action
------	--------

- | | |
|----|--|
| 15 | Select the Server tab to display the following: |
|----|--|



- | | |
|----|---|
| 16 | Verify that PPP: Windows NT, Windows 95 Plus, Internet has been selected from Dial-up server type . |
|----|---|

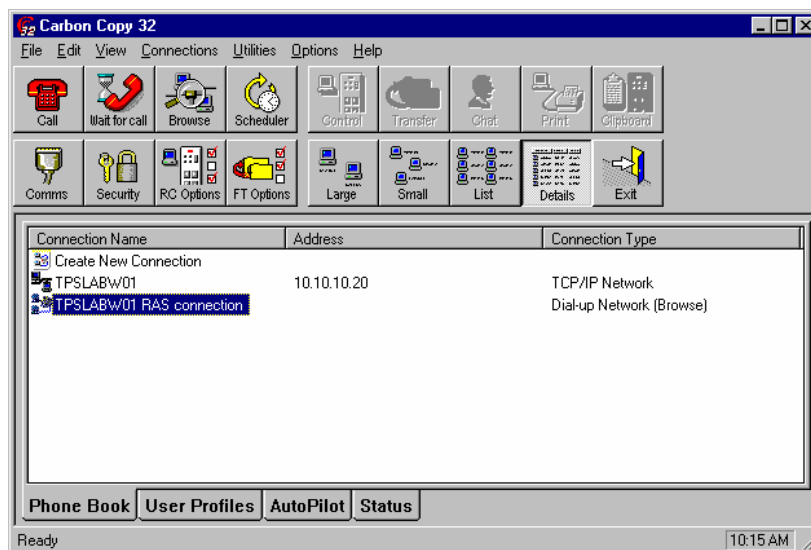
Step	Action
17	Select the Security tab to display the following: 
18	Check the following: Accept any authentication including clear text
19	Click the OK button to complete the connection creation.
20	Continue to the section 4.7 Making a Call.

4.7 Making a Call

- | Step | Action |
|------|---|
| 1 | At the GUS Server, click on the Carbon Copy icon in the task bar tray and select Show User Interface . |
| 2 | Click on Carbon Copy on the task bar to display the user interface. |
| 3 | Select the Waiting button. RESULT: Button changes to Wait for Call . |



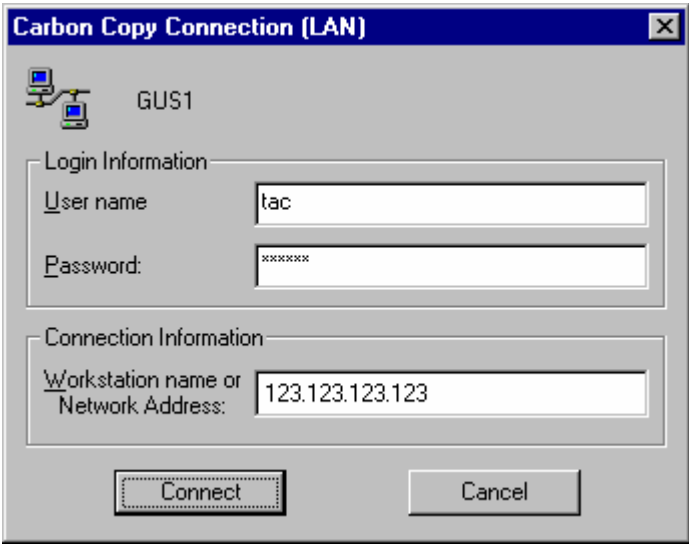
- 4 At the GUS Client, double-click on the calling party's icon in the Phone Book.



RESULT The Carbon Copy Connection dialog box opens.

- 5 Make sure information is correct, and then click **Connect**.

ATTENTION: The Login Name may require domain\username.

Step	Action
	 The image shows a Windows-style dialog box titled "Carbon Copy Connection (LAN)". It has a blue title bar with a close button. Inside, there's a small icon of two computers connected by a line, followed by the text "GUS1". Below this, there are two sections: "Login Information" and "Connection Information". The "Login Information" section has two text boxes: "User name" with the text "tac" and "Password:" with masked text "xxxxxx". The "Connection Information" section has a text box labeled "Workstation name or Network Address:" with the text "123.123.123.123". At the bottom, there are two buttons: "Connect" and "Cancel".
	RESULT: Carbon Copy sends a message to the host stating the "user" (for example "tac" requests access). If access is denied, you are informed. If the connection is approved, the Carbon Copy splash screen is displayed, followed by a window showing the GUS Server (Host) computer screen.

4 Setting Up a Connection

4.7 Making a Call

5. Remote IKB Client Installation and Configuration

5.1 Remote IKB Client Overview

The Remote IKB Client package needs to be installed and configured on any GUS Client containing an IKB (Integrated Keyboard) or OEP (Operator Entry Panel) to support the LED status, horn annunciation, and keylock position during Carbon Copy AE sessions.

Upon system restart, the Remote IKB Client connects to the pre-configured GUS Server (Host) node name. The GUS Server (Host) node name is defined using the IKB Monitor utility.

The IKB Client service can be started and stopped manually at the GUS Client.

Whenever the IKB Client detects loss of communication with the GUS Server (Host), it flashes all of the LEDs on the keyboard and notifies the IKB Monitor utility. Loss of communication is detected within 30-60 seconds.

If the GUS Server (Host) detects loss of communication with a connected IKB Client, it reduces the number of connections by one (effectively freeing up one of the available connections). Loss of communication with an IKB Client does not affect communications with other connected clients.

The IKB Server contains auto-reconnect logic which re-establishes communications when a connection is dropped.

5 Remote IKB Client Installation and Configuration

5.2 Install Remote IKB Client Software



ATTENTION

If you uninstall Remote IKB Client software, then the machine must be rebooted before re-installation can take place.

Step	Action
1	Insert the TPS System Software CD-ROM and select TPS Application Software Packages from the menu of items.
2	Enter the license information, and then select the Remote IKB Client package on the Install dialog page.
3	After software installation, continue to the section 5.3 Configure Remote IKB Client.

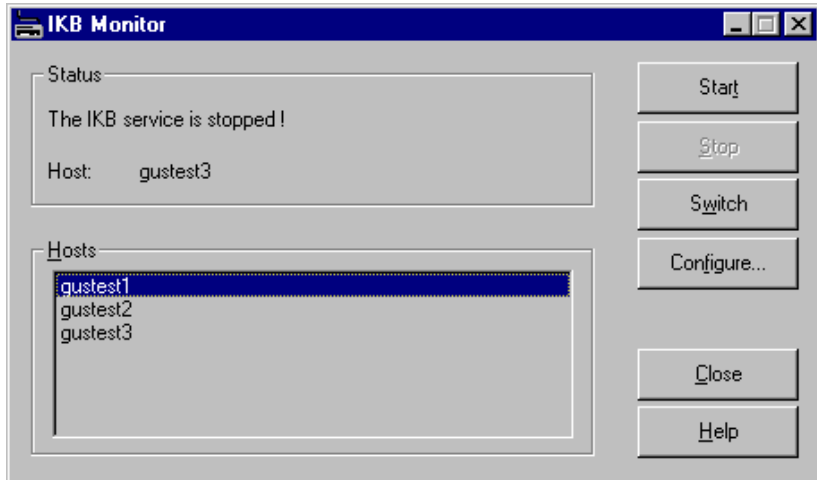
5.3 Configure Remote IKB Client

Purpose

The IKB Monitor utility is a stand-alone, dialog-based application used to configure the GUS Server host names that the Remote IKB Client may connect to.

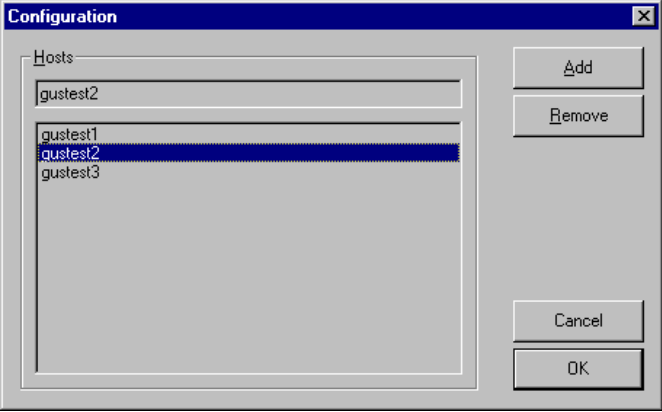
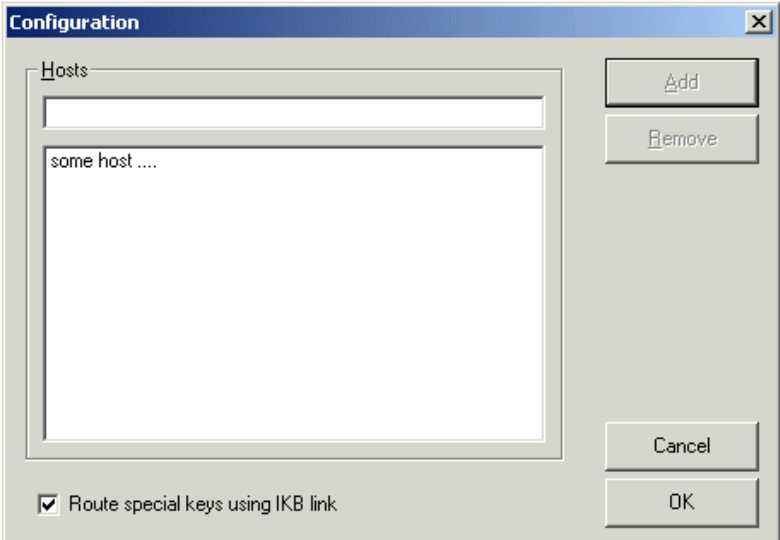
Depending on the user account rights and the configured Windows registry access rights, some of the buttons and fields may be hidden.

Procedure

Step	Action
1	Select the following: Start > Programs > Honeywell TPS > IKB Monitor RESULT: The IKB Monitor dialog box is displayed. 

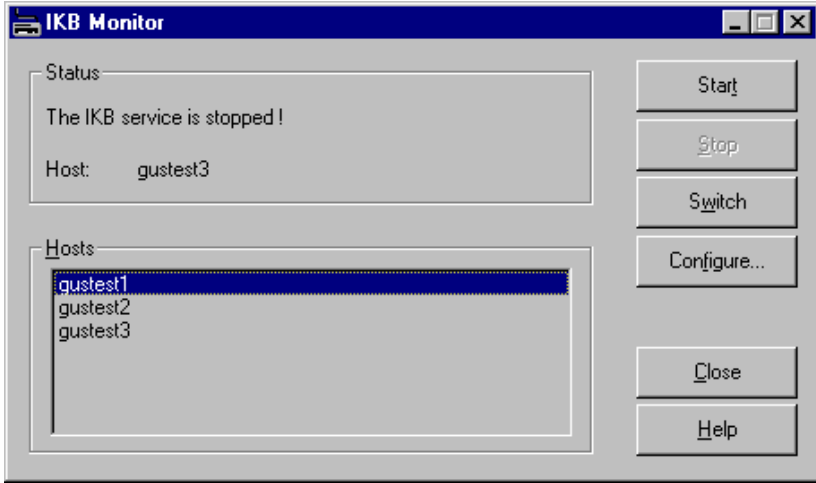
5 Remote IKB Client Installation and Configuration

5.3 Configure Remote IKB Client

Step	Action
2	Select the Configure button. RESULT: The host Configuration dialog box is displayed.  Route special keys using IKB link (GUS R331 and later) • Enable (check) this option on all Remote GUS (Carbon Copy) clients. • The option takes effect when the IKB Service starts. Execute Stop/Start actions on the IKB Monitor page to restart the service. 

5 Remote IKB Client Installation and Configuration

5.3 Configure Remote IKB Client

Step	Action
3	Click Add , specify the host name of the GUS Server, and then click OK .
4	Double-click on the new host name to switch the IKB Client to the new host.  RESULT: The selected host name applies immediately, independent of the state of the IKB Service.
5	Click Start. RESULT: A connection to the host is activated and the IKB Service started.

IKB Monitor dialog box

Dialog Box Item	Description
Status	The Status area displays the current status of the IKB Client connection and the currently assigned GUS Server (Host).
Hosts	The Hosts listbox shows all pre-configured GUS Server host names. The Hosts listbox is hidden if the number of configured GUS Server hosts is 1 or if access rights to the Windows registry are insufficient. Double-click on a host name to switch the IKB Client to the new host. The host name persists through system restart.
Start	Select the Start button to activate a connection to the current host and start the

5 Remote IKB Client Installation and Configuration

5.3 Configure Remote IKB Client

Dialog Box Item	Description
	IKB Service. ATTENTION: Only operators that belong to the Windows Power User or Administrators group may start/stop and select IKB connection to the host.
Stop	Select the Stop button to disconnect from the current host and stop the IKB Service. This Stops any attempts to (re)connect to the host. ATTENTION: Only operators that belong to the Windows Power User or Administrators group may start/stop and select IKB connection to the host.
Switch	The Switch button is hidden if number of configured GUS Server hosts is 1 or if access rights to the Windows registry are insufficient. Select the Switch button to switch the IKB Client to the host selected in the Hosts listbox. The selected host name applies immediately, independent of the state of the IKB Service. The host name persists through system restart.
Configure	The Configure button displays the host Configuration dialog. The Configure button is hidden if access rights to the Windows registry are insufficient. Route special keys using IKB link: When checked, this option enables routing of the IKB "special" keys directly to the GUS Server (Host), bypassing third-party remoting agents (such as Carbon Copy). The "special" keys (such as Alarm Summary) will be sent to the host, independent of which window has focus on the client PC. • This option should be checked for all Remote GUS (Carbon Copy) clients. • Uncheck this option when not using Carbon Copy or any other third-party remoting agent.
Close	The Close button closes the dialog and exits the IKB Monitor utility.
Help	The Help button displays IKB Monitor utility help topics.

6. Remote GUS (Carbon Copy) Operation

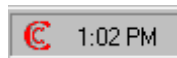
6.1 Carbon Copy AE User Interface

Invoking the user interface

Select the following menu item to invoke Carbon Copy AE:

Start>Programs>Carbon Copy Access Edition>Carbon Copy

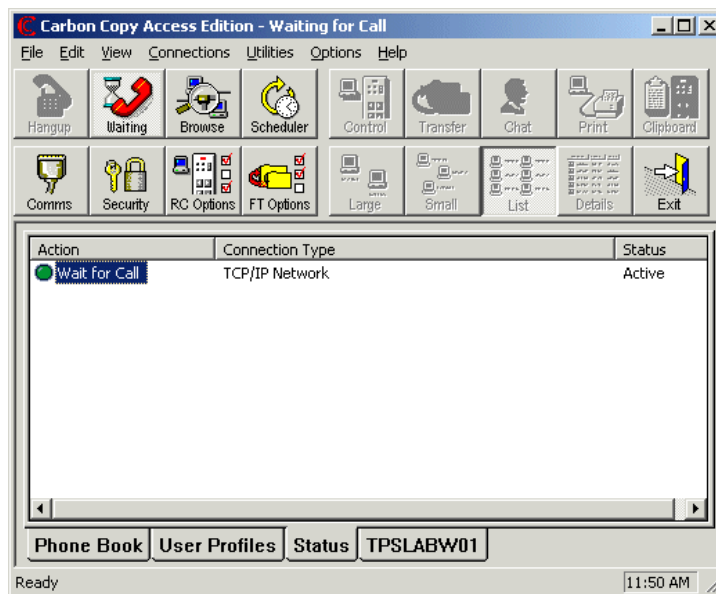
Once the application is active, double-click the Carbon Copy icon in the Windows task bar tray to open the application.



User tasks

The Carbon Copy AE user interface (shown below) is used to do the following:

- configure options (such as communication, security, and startup),
- enter/exit the "wait-for-call" state at the host,
- connect/disconnect to a remote host, and
- invoke utilities, such as Remote Control and File Transfer.



6.2 Carbon Copy AE Functions

Remote Control function

The Remote Control function will do the following:

- Present the GUS Server (Host) screen on the GUS Client screen
- Send mouse events to the GUS Server (Host) if the Remote Control window has focus on the GUS Client
- Send keyboard events to the GUS Server (Host) if the Remote Control window has focus on the GUS Client.

Carbon Copy AE is configured on GUS Clients to start the Remote Control utility automatically upon connection to the GUS Server (Host).

File Transfer function

In order to do bi-directional file transfer between the GUS Server (Host) and the GUS Client, the File Transfer utility must be used.

Emulated Bernoulli drives map into files on the GUS Server (Host) local drive.

Remote Printing function

The Remote Printing function allows the user to print a GUS Server (Host) file at a printer that is connected to the GUS Client.

The Remote Printing Function is used when no shared printer exists.

Chat function

The Chat function allows the user to talk to other Carbon Copy AE users who are connected to the GUS Server (Host). Note that this requires additional hardware (sound card, speakers, microphone) on each user's computer.

Remote Clipboard function

The Remote Clipboard function establishes a common clipboard between the GUS Client and the GUS Server (Host). This is useful when GUS Display Builder sessions are concurrently running on both the GUS Client and GUS Server (Host).

6.3 General Carbon Copy AE Usage Steps

Step	Action
1	The Gus Server (Host) enters the "wait-for-call" state. Carbon Copy AE Service - The Carbon Copy AE service must be running on the GUS Client before the "wait-for-call" state can be entered. The service can be started manually through the Start menu, or the service can be started automatically after each system restart by selecting the following menu item: Options/Security/Startup The "wait-for-call" State - The "wait-for-call" state can be entered either manually through the Carbon Copy AE user interface, or the "wait-for-call" state can be started automatically after each system restart by selecting the following menu item: Options/Communications/Wait For Call/Options
2	The GUS Client user activates a connection to the GUS Server (Host), usually by using one of the predefined Carbon Copy AE "Phone Book" connections.
3	The GUS Server (Host) accepts the connection. The GUS Server (Host) Carbon Copy AE security settings might require that a local operator at the GUS Server (Host) acknowledge the connection request by the GUS Client. The security settings might also require the GUS Client user to log on to the GUS Server.
4	The GUS Client user invokes the needed Carbon Copy AE utilities and conducts the remote session. Carbon Copy AE utilities can be configured to start automatically when the GUS Client connects to the GUS Server (Host).
5	The GUS Client user disconnects from the GUS Server (Host).



ATTENTION

The Carbon Copy AE communications link has the following limitations:

- The GUS Server (Host) can only service one GUS Client at a time.
 - The GUS Client may predefine the number of GUS Servers (hosts) it intends to connect to, but only one active connection to a GUS Server (Host) can exist at a time.
-

6.4 Remote GUS Screen Usage

Screen Usage on the GUS Server (Host)

The screen should be configured as recommended by the TPS Display Authoring Tutorial - GUS User Interface Styles.

In remote operations scenarios, a single screen is supported.

Single-Screen Usage on the GUS Client

The Carbon Copy AE Remote Control function maintains a window on the GUS Client that displays the contents of the entire GUS Server (Host) desktop. The Remote Control window option is dependent upon whether you will use SafeView:

- If you are going to use SafeView, then do not use the Remote Control "full screen" option.
- If you are not going to use SafeView, then you may use the Remote Control "full screen" option if the screen resolution of the GUS Client and the GUS Server (Host) is identical. In this mode, the whole GUS Client screen presents the entire desktop of the GUS Server (Host), and a keyboard key combination is used to access the desktop on the GUS Client.

6.5 Remote Usage Recommendations

Area	Recommendation
Connectivity	Carbon Copy AE Internet connectivity is not supported.
GUS Display Builder Animation	Animation is not recommended.
GUS Display Builder Color	Internally, Carbon Copy AE uses a 256 color-mapping algorithm. For this reason, color fidelity may be affected at the GUS Client.
GUS Display Builder Operator Input	Keystrokes may be lost between a mouse click (invoking a user input script) and when the data edit control appears on the screen. Therefore, Honeywell recommends that data modifications be confirmed before being written to the process variable.
GUS Display Builder Performance	Do not operate more than two GUS Displays simultaneously. Do not use update rates faster than 4 seconds.
Middle roll/button	Even with the appropriate mouse driver present, Carbon Copy does not emulate the middle roll/button.
Language	Carbon Copy AE is only provided in an English version.
Printer	The "console" (Native Window) printer may not be redirected to the printer at the remote GUS Client.
Remote Native Window	Concurrent operation of the Remote Native Window and Carbon Copy AE is not supported.
Multiple Remote Control Sessions	Multiple concurrent sessions of the Carbon Copy AE Remote Control at the GUS Client are not recommended. This is for performance reasons and to avoid possibly dangerous situations where the Remote IKB Client and the current Remote Control window of focus point to different GUS Servers (Hosts).

6.6 Examples of Remote GUS User Roles

Remote operations user roles

Remote operators connecting to a GUS Server (Host) may serve in any one of a number of operation roles, including (but not limited to) the following roles described in this section:

- remote console operators
- roving operators
- remote maintenance operators

Remote console operators

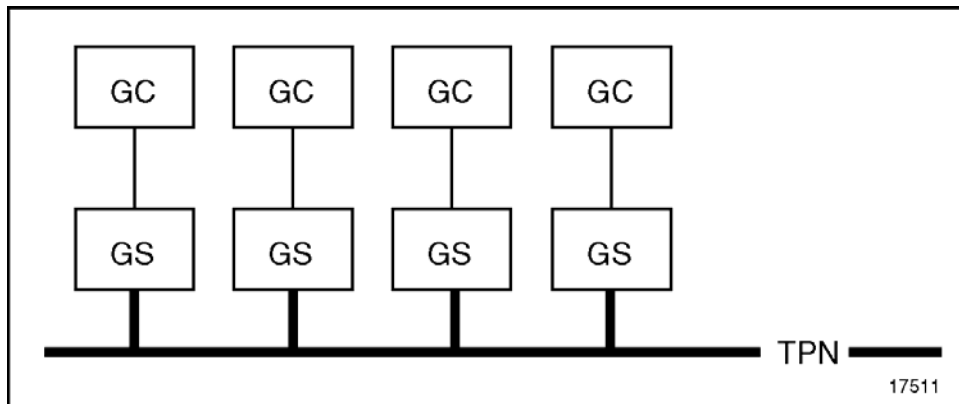
"Remote console operators" use multiple GUS Clients and GUS Servers (Hosts) to view into the controlled process. The remote console topology is static and is depicted in the following illustration.



ATTENTION

Note that GC represents the GUS Client, and GS represents GUS Server.

Remote Console Operations Topology



Roving operator

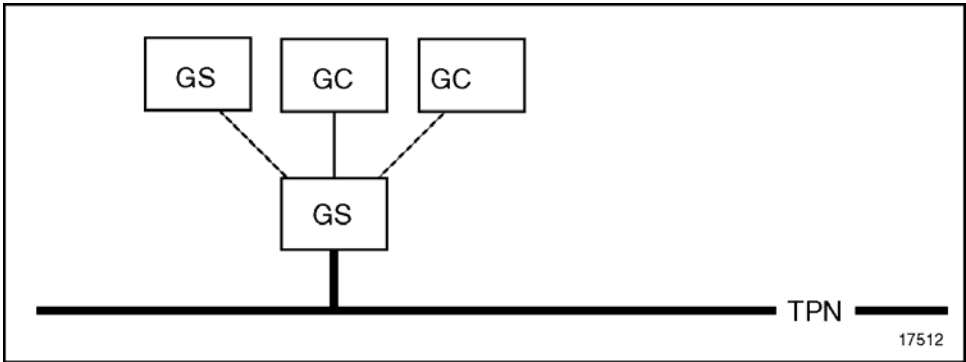
A "roving operator" moves between multiple GUS Clients, which are connected to the same GUS Server (Host). The GUS Server can service only one GUS Client at a time. The following illustration depicts the roving operator topology.



ATTENTION

Note that GC represents the GUS Client, and GS represents GUS Server.

Roving Operator Topology



6 Remote GUS (Carbon Copy) Operation

6.6 Examples of Remote GUS User Roles

Remote maintenance operators

"Remote maintenance operators" connect to GUS Servers (Hosts), which are located on different sites, connecting to one GUS Server at a time.

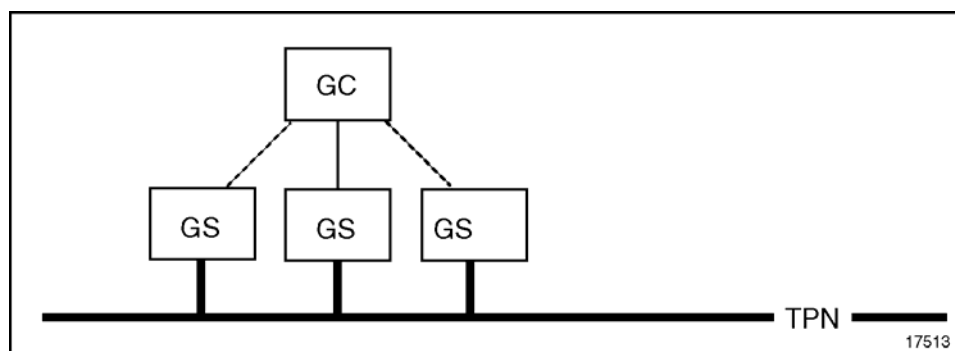
Operator actions are usually limited to verifying settings and occasional changes of the process variables. The following illustration depicts the remote maintenance topology.



ATTENTION

Note that GC represents the GUS Client, and GS represents GUS Server.

Remote Maintenance Topology



Remote engineering user roles

An engineer/expert connecting from the remote GUS Client may assist the local operator.

A "remote engineer" can do tasks, such as the following:

- interpret runtime error information,
- provide technical assistance,
- review and modify GUS Display scripting,
- verify GUS Server (Host) configuration settings,
- develop GUS Displays and CL programs, and
- configure the system (such as Display Builder configuration and CL Server configuration).

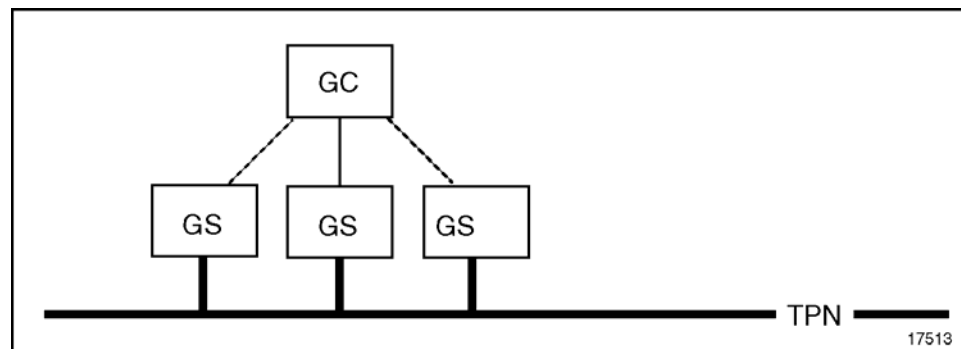
A "remote engineer" typically configures connections to GUS Servers (Hosts) located at different sites, and then uses one connection at a time, as depicted in the following figure.



ATTENTION

Note that GC represents the GUS Client, and GS represents GUS Server.

Remote Engineering Topology



6.7 Remote Usage Scenarios

Connect GUS Client to GUS Server (Host)

Step	Action
1	The user logs on to the GUS Client machine with their predefined GUS Client user account.
2	The user invokes Carbon Copy AE to connect to the GUS Server (Host). RESULT: The Carbon Copy AE Remote Control utility automatically starts and opens the GUS Server (Host) "Remote Control" window on a GUS Client screen.
3	The user gives the focus to the "Remote Control" window.
4	The user logs on to the GUS Server (Host) with their predefined GUS Server (Host) user account.
5	The user activates the GUS and Native Window displays and operates the site.

Restart GUS Server (Host) after a power failure

Step	Action
1	The GUS Server (Host) experiences a power failure.
2	The GUS Server (Host) reboots.
3	The Carbon Copy AE components on the GUS Server (Host) automatically enter the "Wait-for-Call" state.
4	If the GUS Server (Host) restarted while the connection was active (for example, if the GUS Server (Host) system Shutdown was initiated by the GUS Client user), then the GUS Client user needs to reconnect to the GUS Server (Host). For detailed instructions, refer to section 6.7 Connect GUS Client to GUS Server (Host).
5	The GUS Client user starts the Native Window and loads the LCNP board from within the "Remote Control" window.
6	The GUS Client user activates the GUS and Native Window displays and operates the site.

Restart GUS Client after a power failure

Step	Action
1	The GUS Client experiences a power failure.
2	The GUS Client reboots.
3	The GUS Client user logs on to the GUS Client with their predefined GUS Client user account.
4	The user invokes Carbon Copy AE to connect to the GUS Server (Host). RESULT: The Carbon Copy AE Remote Control utility automatically starts and opens the GUS Server (Host) "Remote Control" window on a GUS Client screen.
5	The GUS Client user gives the focus to the "Remote Control" window.
6	The GUS Client user logs on to the GUS Server (Host) with their predefined GUS Server (Host) user account.
7	The GUS Client user activates the GUS and Native Window displays and operates the site.

Restart GUS Server (Host) after a Carbon Copy AE failure

Step	Action
1	The GUS Client user attempts to connect to a GUS Server (Host) that is part of a TPS domain.
2	They receive a connection error message.
3	They notify the System Administrator who has TPS domain administrator privileges.
4	The System Administrator remotely reboots the GUS Server (Host) using the Carbon Copy restart utility
5	The GUS Client user logs on to the GUS Client with their predefined GUS Client user account.

6 Remote GUS (Carbon Copy) Operation

6.7 Remote Usage Scenarios

Step	Action
6	The GUS Client user invokes Carbon Copy AE to connect to the GUS Server (Host). RESULT: The Carbon Copy AE Remote Control utility automatically starts and opens the GUS Server (Host) "Remote Control" window on a GUS Client screen.
7	The GUS Client user gives the focus to the "Remote Control" window.
8	The GUS Client user logs on to the GUS Server (Host) with their predefined GUS Server (Host) user account.
9	The GUS Client user activates the GUS and Native Window displays and operates the site.

Use a GUS Client with an IKB or OEP

Step	Action
1	The user logs on to the GUS Client with a pre-configured connection to a GUS Server (Host).
2	The user uses the Remote IKB Client Monitor utility to activate the Remote IKB Client.
3	The user uses Carbon Copy AE and a predefined user account to connect to the GUS Server (Host). RESULT: The Carbon Copy AE Remote Control utility automatically starts and opens the GUS Server (Host) "Remote Control" window on a GUS Client screen.
4	The user gives the focus to the "Remote Control" window.
5	The user logs on to the GUS Server (Host) with their predefined GUS Server (Host) user account.
6	The user activates GUS and Native Window displays and operates the site.

Step	Action
7	To end the session, the user does the following: 1.) The user logs off from the GUS Server (Host). 2.) The user terminates the Carbon Copy AE "Remote Control" utility. 3.) The user uses the Remote IKB Client Monitor utility to stop the Remote IKB Client. 4.) The user logs off of the GUS Client.

Terminate another connected GUS Client

The following scenario will work only if

- Carbon Copy AE Host components are active on the GUS Client,
- GUS Client is in the "wait-for-call" state, and
- GUS Client is configured for "unmanned" operation.

Step	Action
1	The user logs on to the GUS Client with a pre-configured connection to a GUS Server.
2	The user attempts to activate Remote IKB and Carbon Copy AE connections to a GUS Server.
3	The user discovers that the GUS Server (Host) is busy servicing another GUS Client.
4	The system returns an error that does not indicate which GUS Client the GUS Server (Host) is currently connected to.
5	The user uses Carbon Copy AE to scan existing GUS Clients until they find the GUS Client connected to the GUS Server.
6	The user uses the Carbon Copy AE "Remote Control" utility to terminate all of the GUS Client connections to the GUS Server.
7	The user terminates their Carbon Copy AE connection to the GUS Client and connects to the GUS Server.

6 Remote GUS (Carbon Copy) Operation

6.7 Remote Usage Scenarios



ATTENTION

This scenario may be implemented in a simpler way by configuring the GUS Server (Host) to accept more than one connection at a time. However, Honeywell recommends a single active connection between the GUS Client and GUS Server (Host) for security and operational reasons.

Perform remote maintenance

Step	Action
1	The remote maintenance operator determines which GUS Server they wish to connect to.
2	The remote maintenance operator uses the Remote IKB Client Monitor utility to select the required GUS Server (Host) and starts the Remote IKB Client.
3	The remote maintenance operator logs on to the GUS Client with their predefined GUS Client user account.
4	The remote maintenance operator uses Carbon Copy AE and a predefined user account to connect to the GUS Server (Host). The Carbon Copy AE Remote Control utility will automatically start and open the GUS Server (Host) "Remote Control" window on a GUS Client screen.
5	The remote maintenance operator gives the focus to the "Remote Control" window.
6	The remote maintenance operator logs in to the GUS Server (Host) with their predefined GUS Server (Host) user account.
7	The remote maintenance operator activates the GUS and Native Window displays and performs maintenance operations at the site.
8	To end the session, the user does the following: 1.) The user terminates the Carbon Copy AE "Remote Control" utility. 2.) The user terminates the Carbon Copy AE connection. 3.) The user uses the Remote IKB Client Monitor utility to stop the Remote IKB Client. 4.) The user logs off of the GUS Client.

Restart GUS Server (Host) after a Carbon Copy AE failure

Step	Action
1	The user attempts to connect to a GUS Server (Host).
2	They receive a connection error message.
3	The user uses a third-party Remote Site Management software/hardware to remotely boot the GUS Server (Host).
4	The user logs on to the GUS Client with their predefined GUS Client user account.
5	The user invokes Carbon Copy AE to connect to the GUS Server (Host). RESULT: The Carbon Copy AE Remote Control utility automatically starts and opens the GUS Server (Host) "Remote Control" window on a GUS Client screen.
6	The user gives the focus to the "Remote Control" window.
7	The user logs in to the GUS Server (Host) with their predefined GUS Server (Host) user account.
8	The user activates the GUS and Native Window displays and operates the site.

Develop GUS displays

Step	Action
1	The remote engineer builds GUS displays using GUS Offline Display Builder.
2	The remote engineer connects to the GUS Server (Host) using Carbon Copy AE.
3	The GUS Server (Host) local operator acknowledges the connection.
4	If GUS Server (Host) and GUS Client are on different networks or if they are connected through modem, then the remote engineer starts the File Transfer utility and transfers GUS Displays to the GUS Server (Host) location.
5	The remote engineer uses the remote control capabilities of Carbon Copy AE to validate and test the displays.
6	The remote engineer modifies the displays.
7	The remote engineer installs operational GUS displays according to local policy.

6 Remote GUS (Carbon Copy) Operation

6.7 Remote Usage Scenarios

Step	Action
8	The remote engineer terminates the Carbon Copy AE connection.

Remote engineer receives remote assistance from TAC

Step	Action
1	An Administrator at the GUS Server (Host) site informs a Honeywell TAC expert about the Windows user credentials needed for the connection.
2	The Administrator at the GUS Server (Host) site starts the Carbon Copy AE service.
3	The local operator at the GUS Server (Host) activates "Wait For Call."
4	The TAC expert connects.
5	The local operator acknowledges the connect request.
6	The TAC expert executes Remote Control and other Carbon Copy AE utilities (Chat to talk to the operator, for example).
7	The TAC expert disconnects.
8	The local operator terminates "Wait For Call."
9	If desired, the local administrator at the host changes the TAC expert Windows user credentials.



ATTENTION

- Step 1 may be executed early, and requires administrative or power user privileges at the GUS Server (Host). During this step, a modem may be attached to the pre-configured COM port on the GUS Server (Host) PC ("a single modem on a site").
 - Steps 2-5 may be repeated for multiple calls from TAC.
 - The remote expert may use a generic TAC user account to connect with Carbon Copy AE.
-

Remote assistance from TAC into the remote GUS installation

Step	Action
1	An Administrator at the GUS Server (Host) site informs a Honeywell TAC expert about the Windows user credentials needed for the connection.
2	The Administrator at the GUS Server (Host) site starts the Carbon Copy AE service.
3	The local operator at the GUS Server (Host) activates "Wait For Call."
4	The TAC expert connects.
5	The local operator acknowledges the connect request.
6	The TAC expert executes Remote Control and other Carbon Copy AE utilities (for example, the Chat utility to talk to the operator). This includes imitating local operator actions to connect to the GUS Server (Host) (using Carbon Copy AE on the GUS Client).
7	The TAC expert disconnects.
8	The local operator terminates "Wait For Call."



ATTENTION

A direct TAC connection into the GUS Server (Host) may not be advisable, since the GUS Server (Host) is configured to accept a single connection only. If a direct connection is made, security may be compromised.

6.8 Remote Control Utility Usage

Configuration precaution

The following precaution should be considered when configuring the Remote Control Utility.



ATTENTION

Some Ctrl/key combinations may stop the Remote GUS Client operation. To avoid this problem, do the following:

When configuring Carbon Copy AE Remote Control Hot Keys, leave everything set to the Default (that is, the key combinations begin with the Alt Key).

Remote restart of GUS Server (Host)

If the Remote Control utility is active, the user at the client PC may restart the host PC. The Ctrl/Alt/Del key combination must be sent as a separate command available on the Remote Control window menu /toolbar.

The scenario for the usage of the Remote Control Utility for a remote restart is as follows. The scenario assumes that the Carbon Copy AE link is operative between the GUS Client (Guest) and the GUS Server (Host).

Step	Action
1	The user at the GUS Client starts the Remote Control Utility.
2	The user at the GUS Client selects the Restart command on the Remote Control Utility toolbar.
3	The user at the GUS Client sees the standard Windows Shutdown dialog.
4	The user at the GUS Client selects <u>R</u>estart .
5	The GUS Server (Host) shuts down, and then restarts.

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